

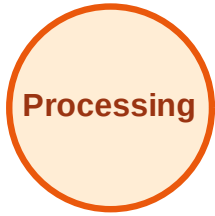
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



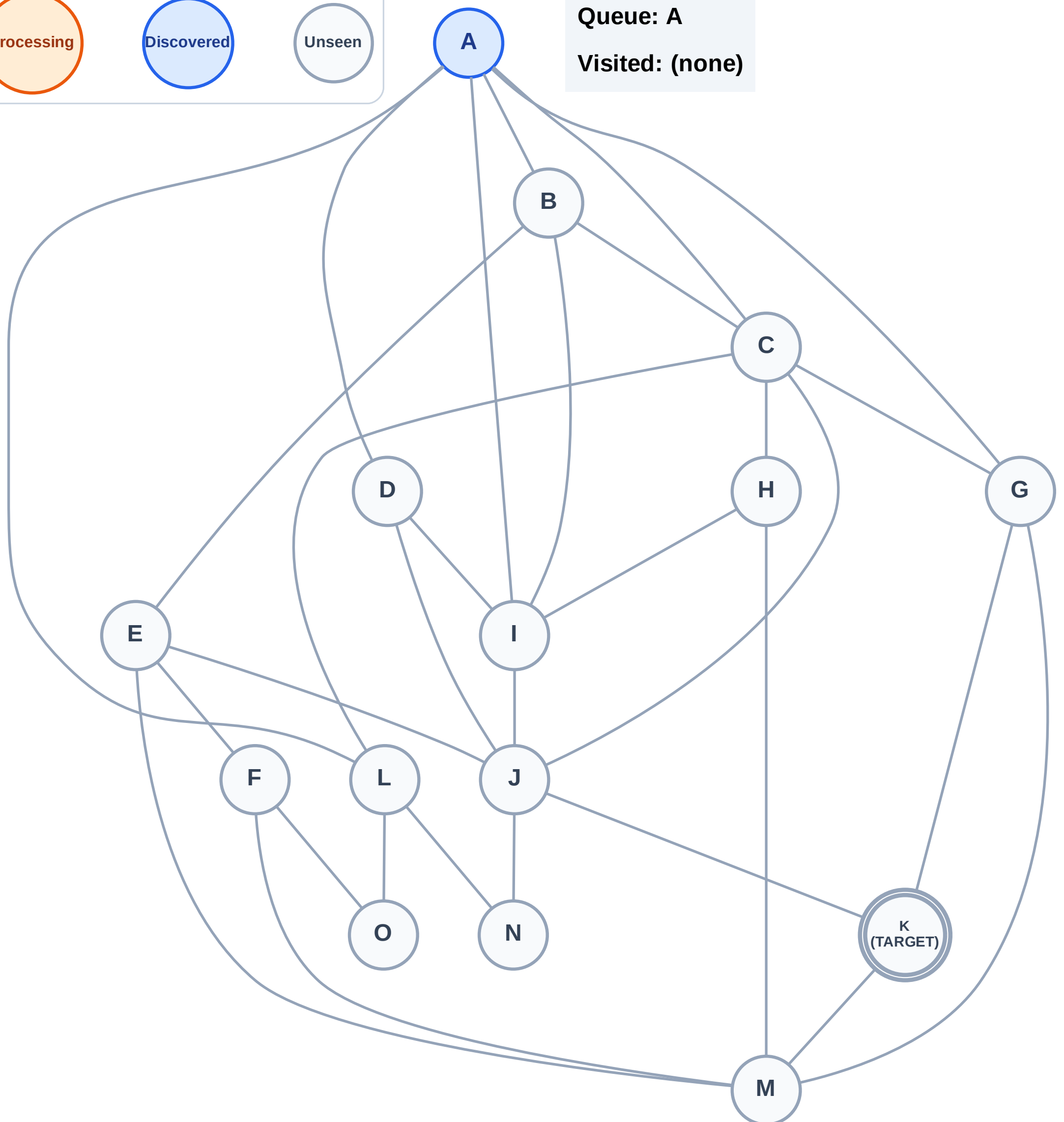
Discovered



Unseen

Queue: A

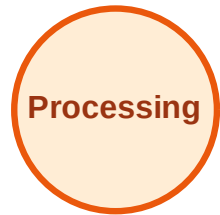
Visited: (none)



BFS Traversal

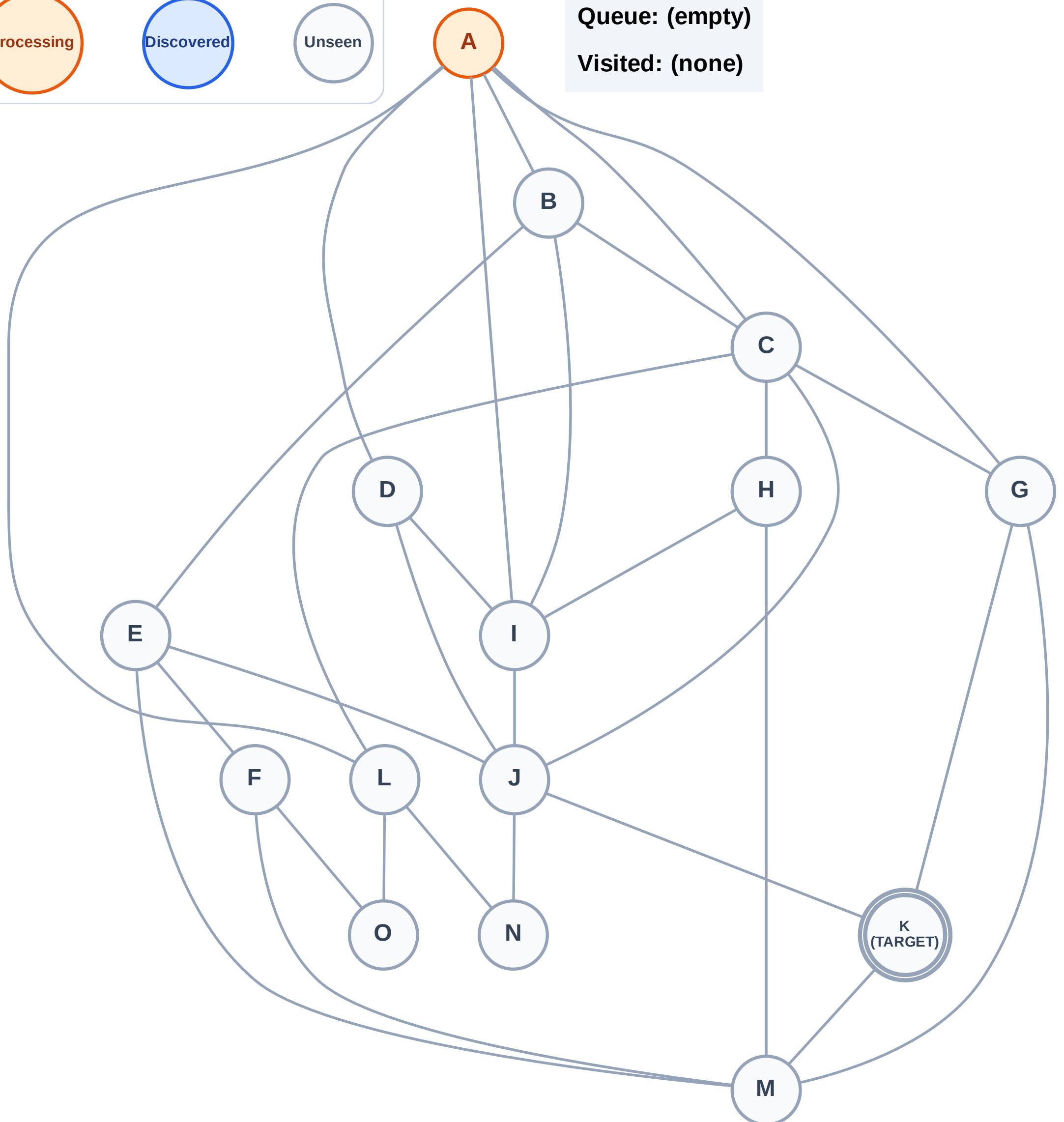
Step 2: Process node A

Legend



Queue: (empty)

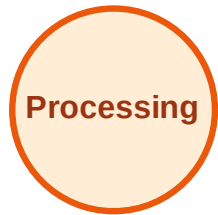
Visited: (none)



BFS Traversal

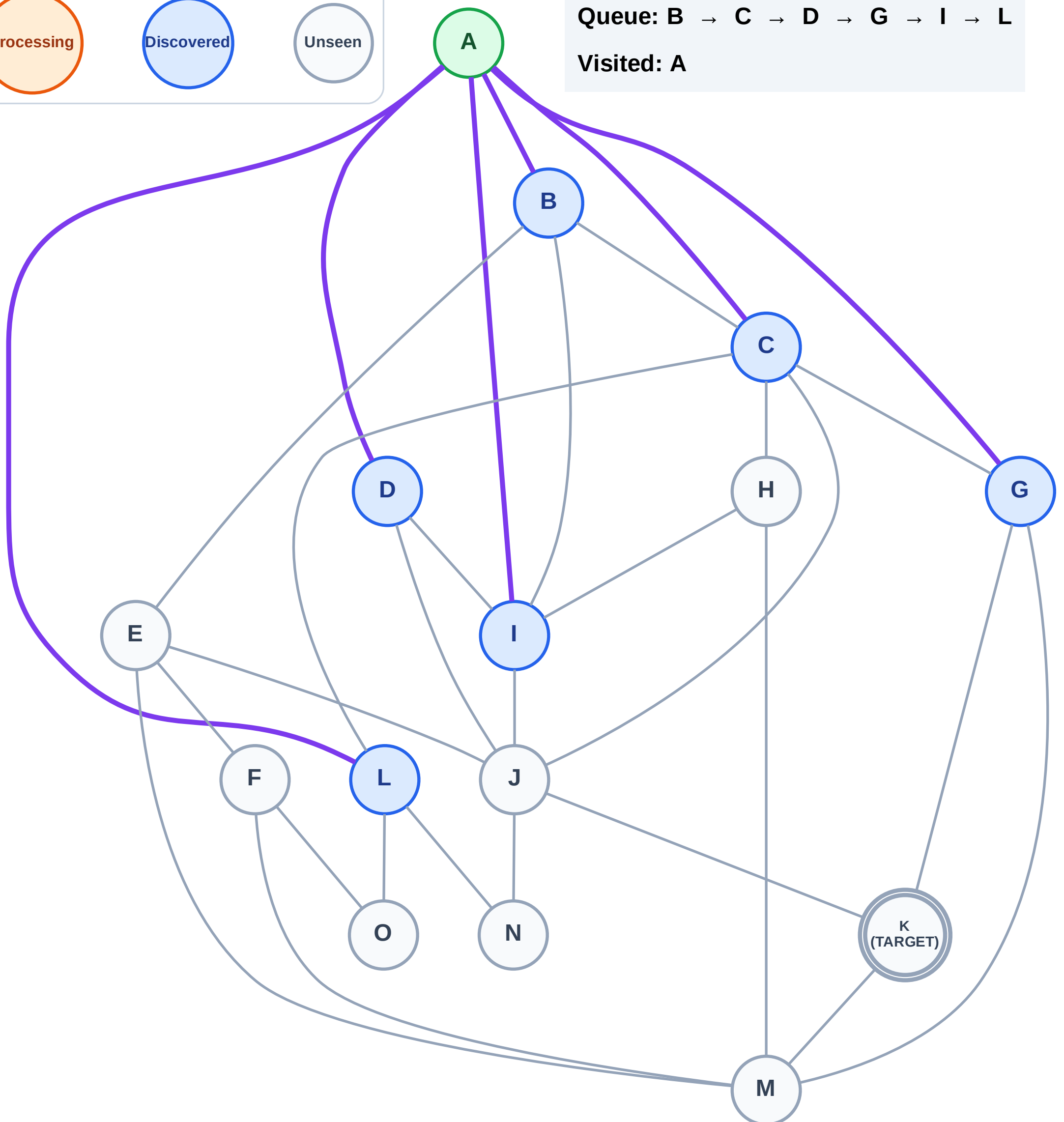
Step 3: Discover B, C, D, G, I, L from A

Legend



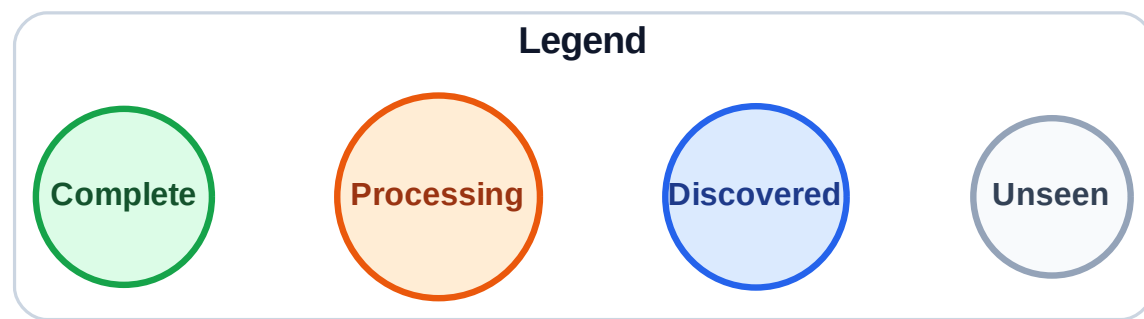
Queue: B → C → D → G → I → L

Visited: A



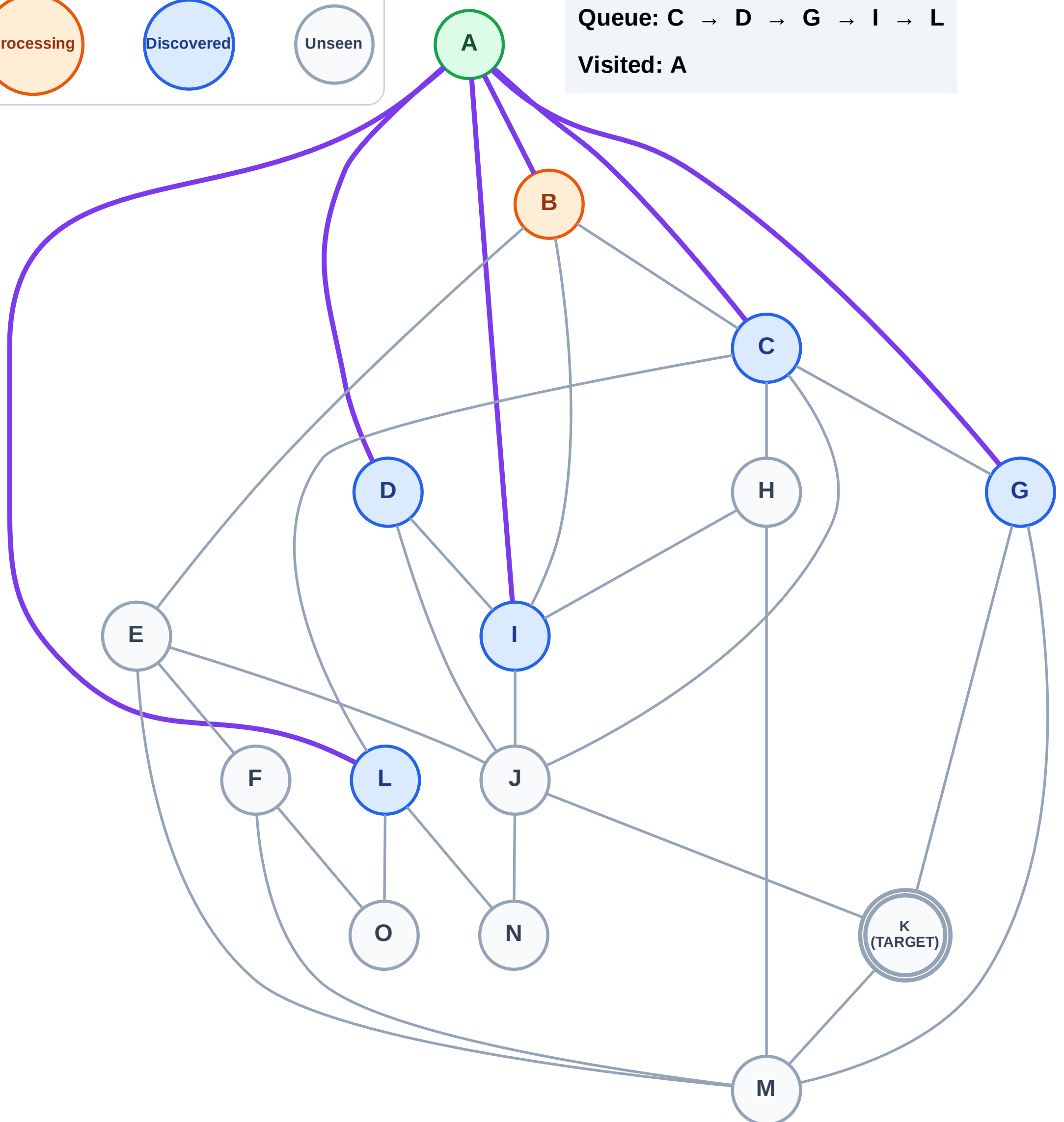
BFS Traversal

Step 4: Process node B



Queue: C → D → G → I → L

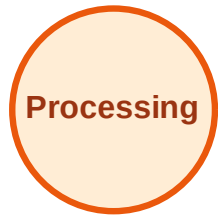
Visited: A



BFS Traversal

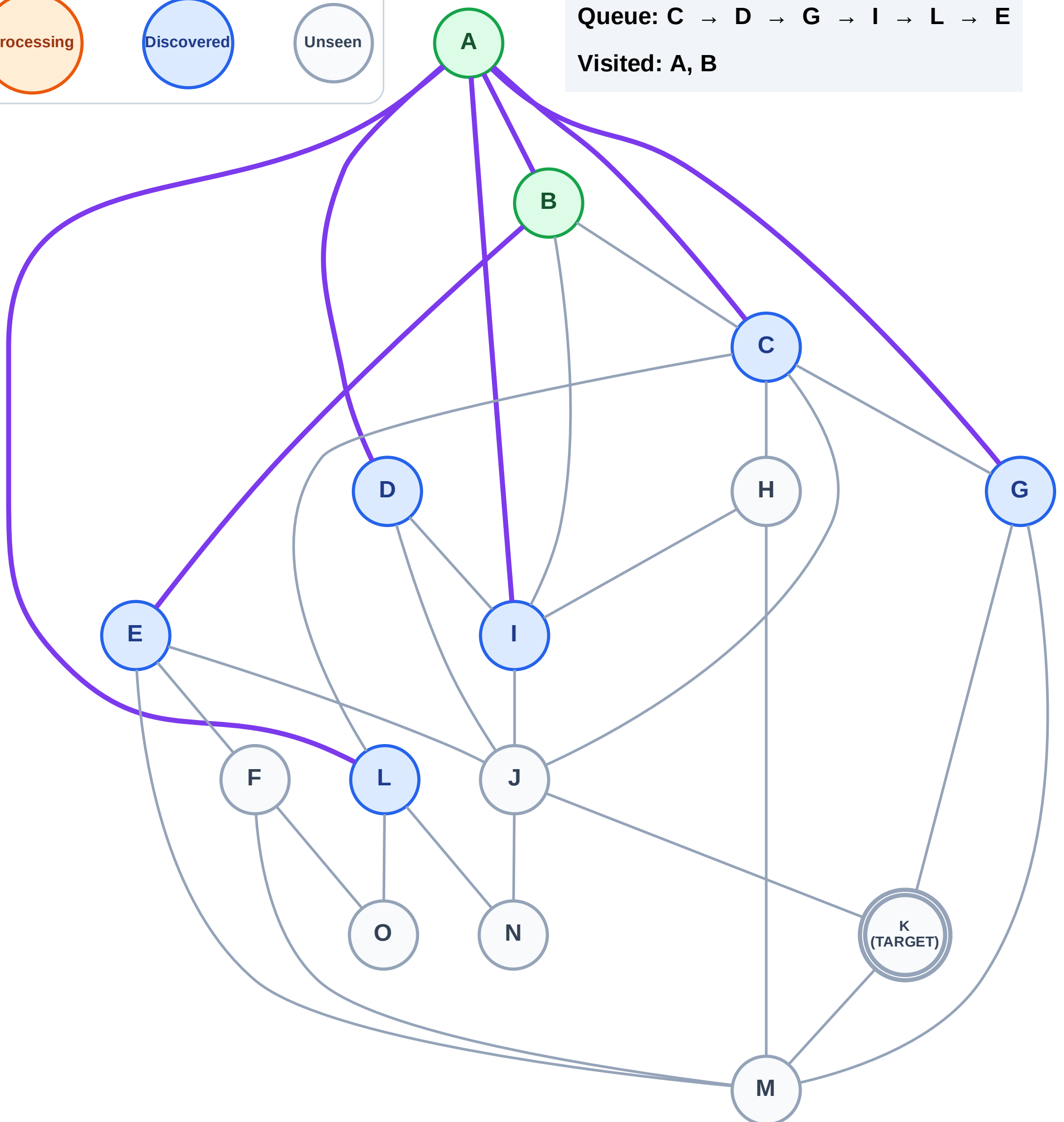
Step 5: Discover E from B

Legend



Queue: C → D → G → I → L → E

Visited: A, B



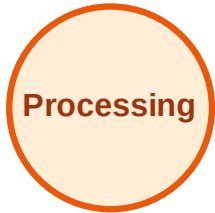
BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



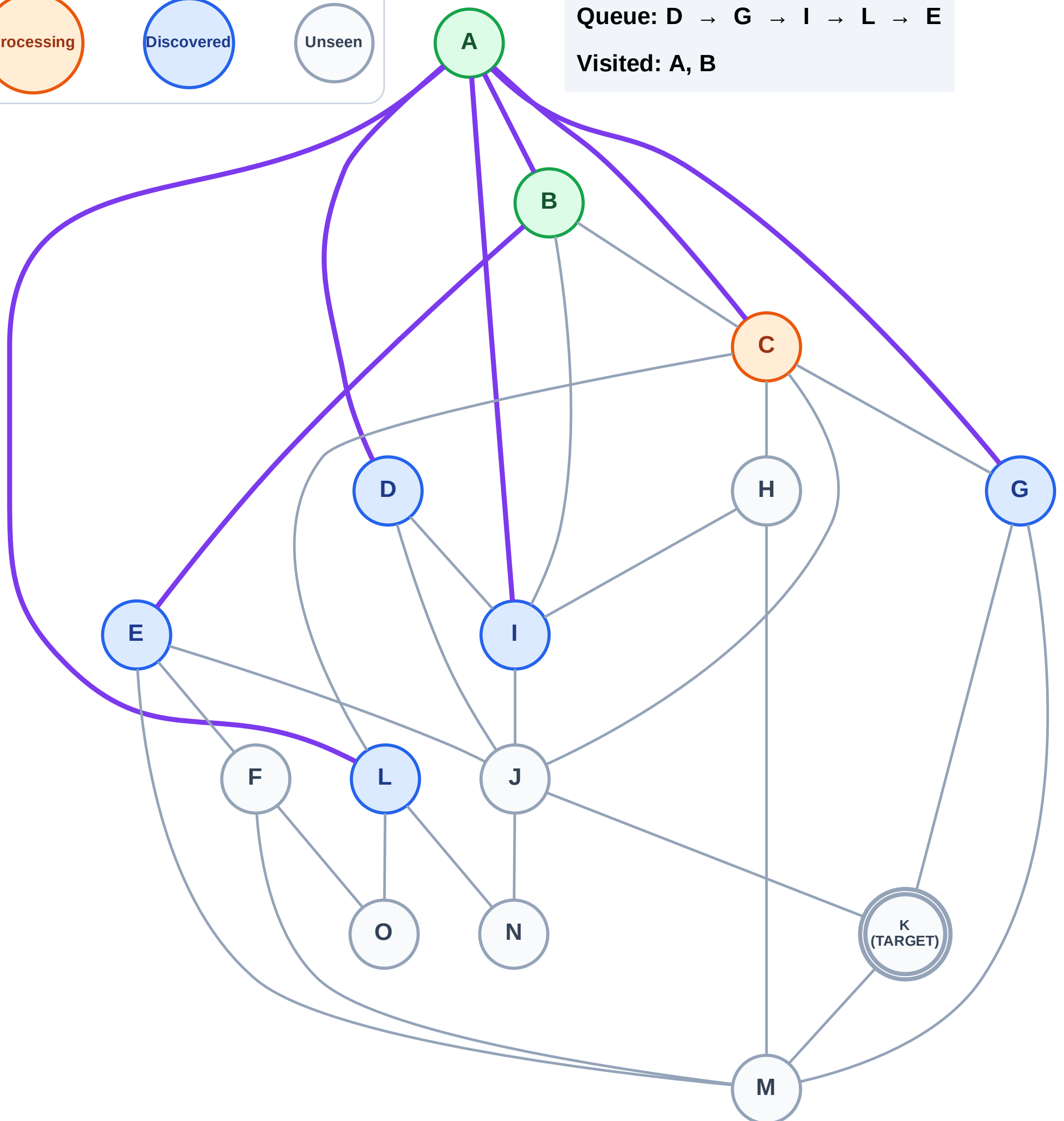
Discovered



Unseen

Queue: D → G → I → L → E

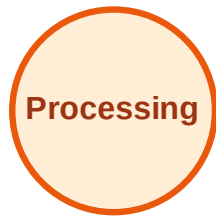
Visited: A, B



BFS Traversal

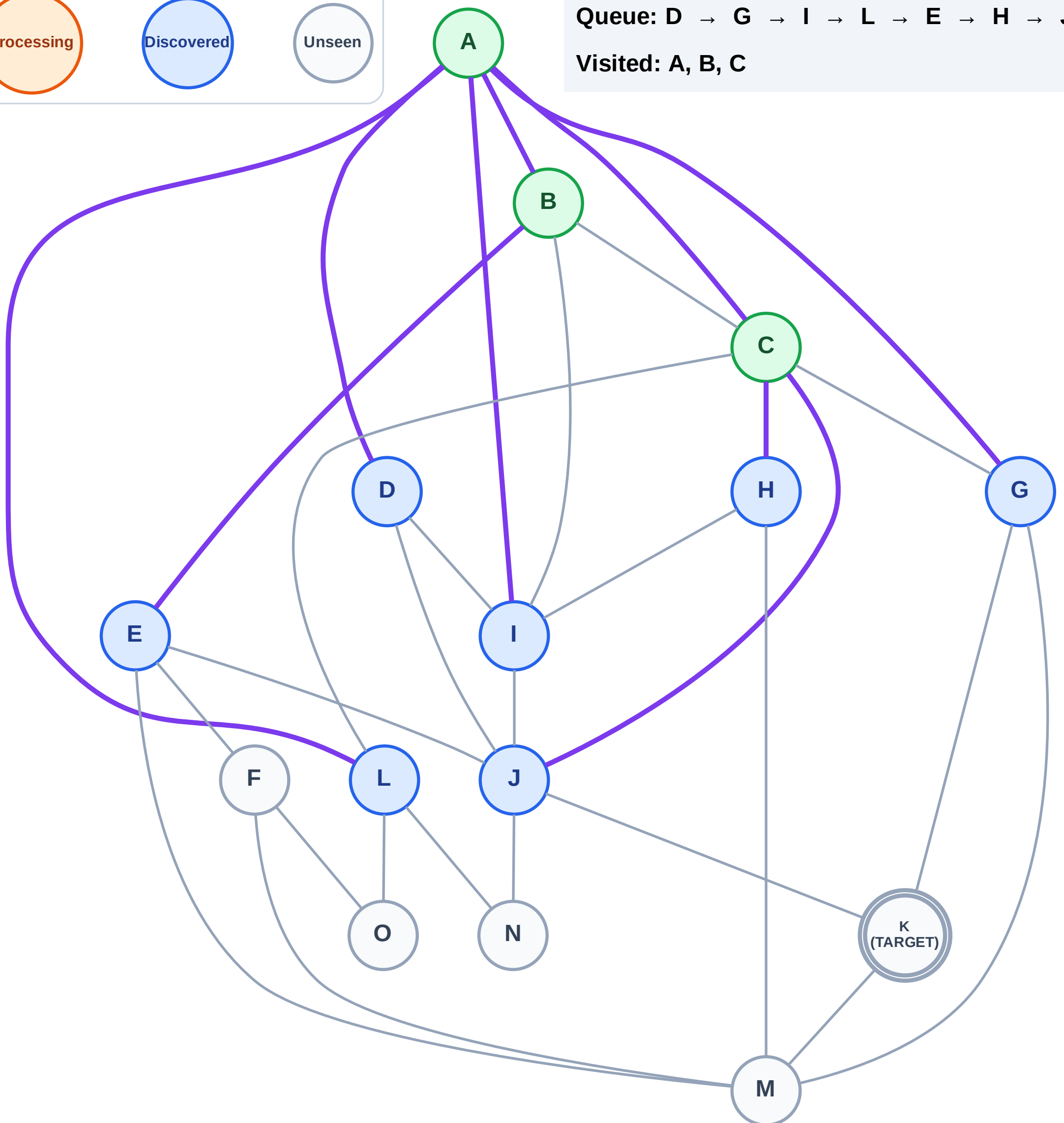
Step 7: Discover H, J from C

Legend



Queue: D → G → I → L → E → H → J

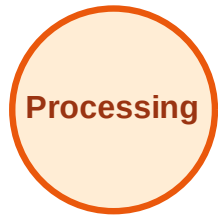
Visited: A, B, C



BFS Traversal

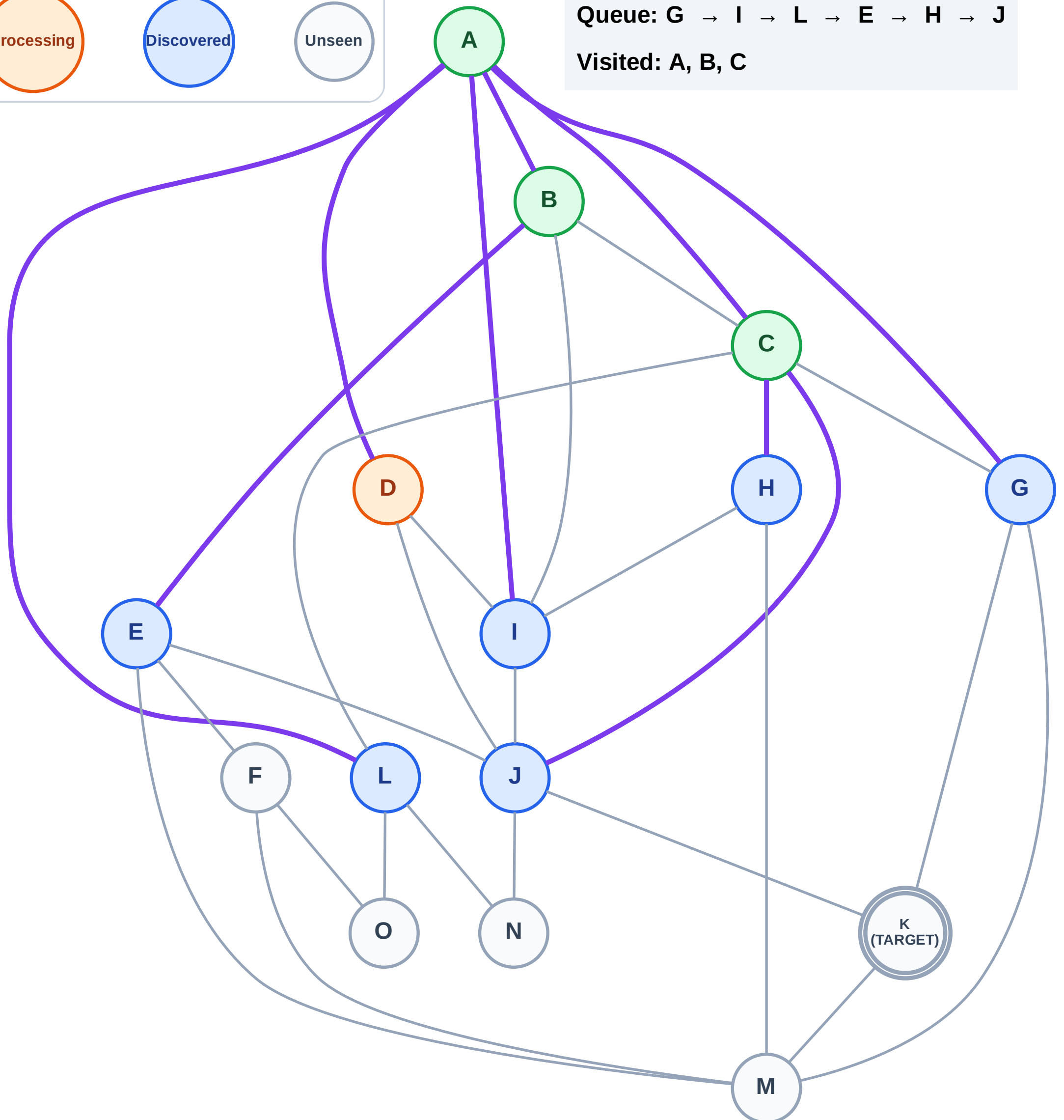
Step 8: Process node D

Legend



Queue: G → I → L → E → H → J

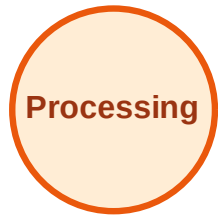
Visited: A, B, C



BFS Traversal

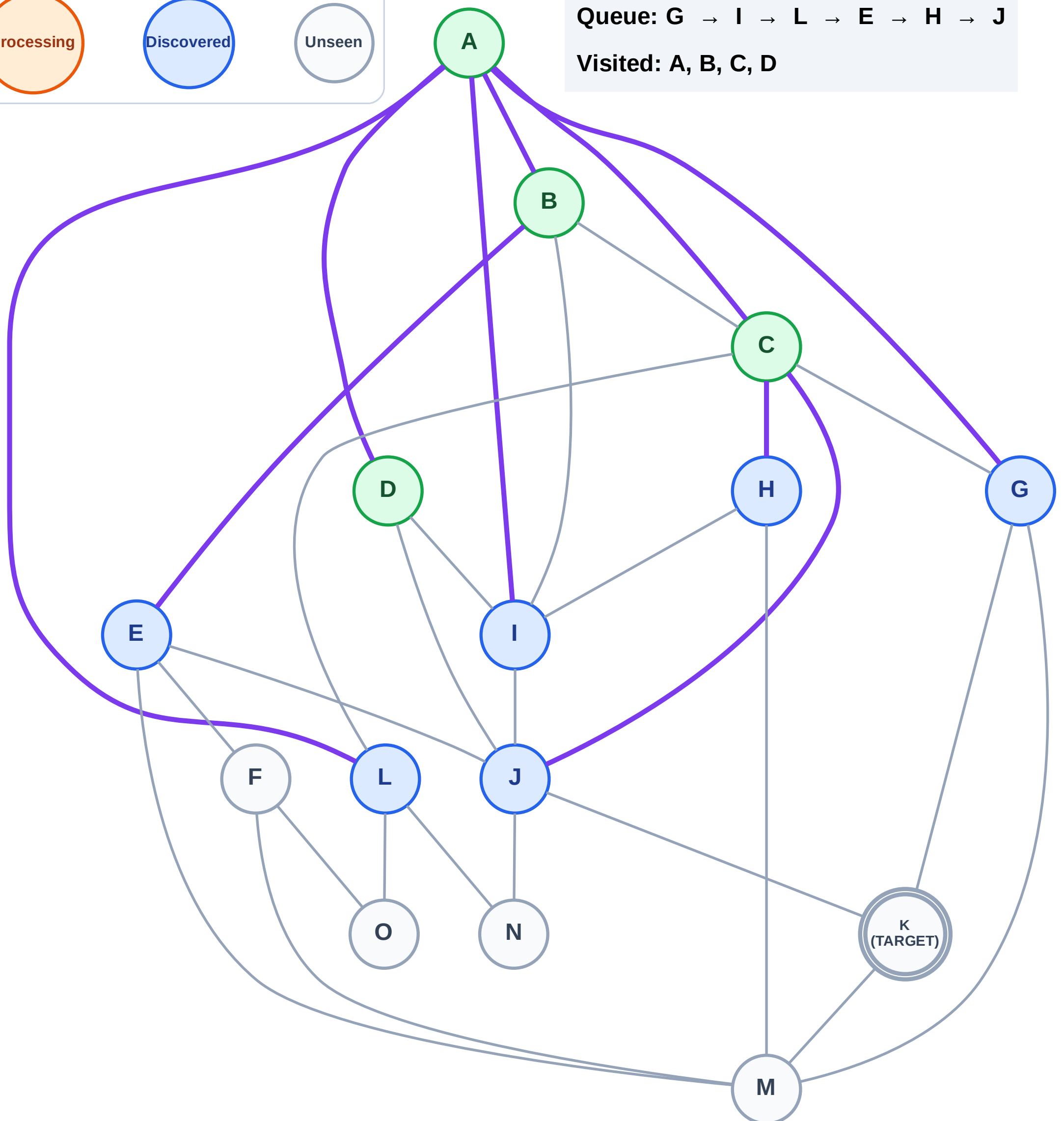
Step 9: No new neighbors from D

Legend



Queue: G → I → L → E → H → J

Visited: A, B, C, D



BFS Traversal

Step 10: Process node G

Legend

Complete

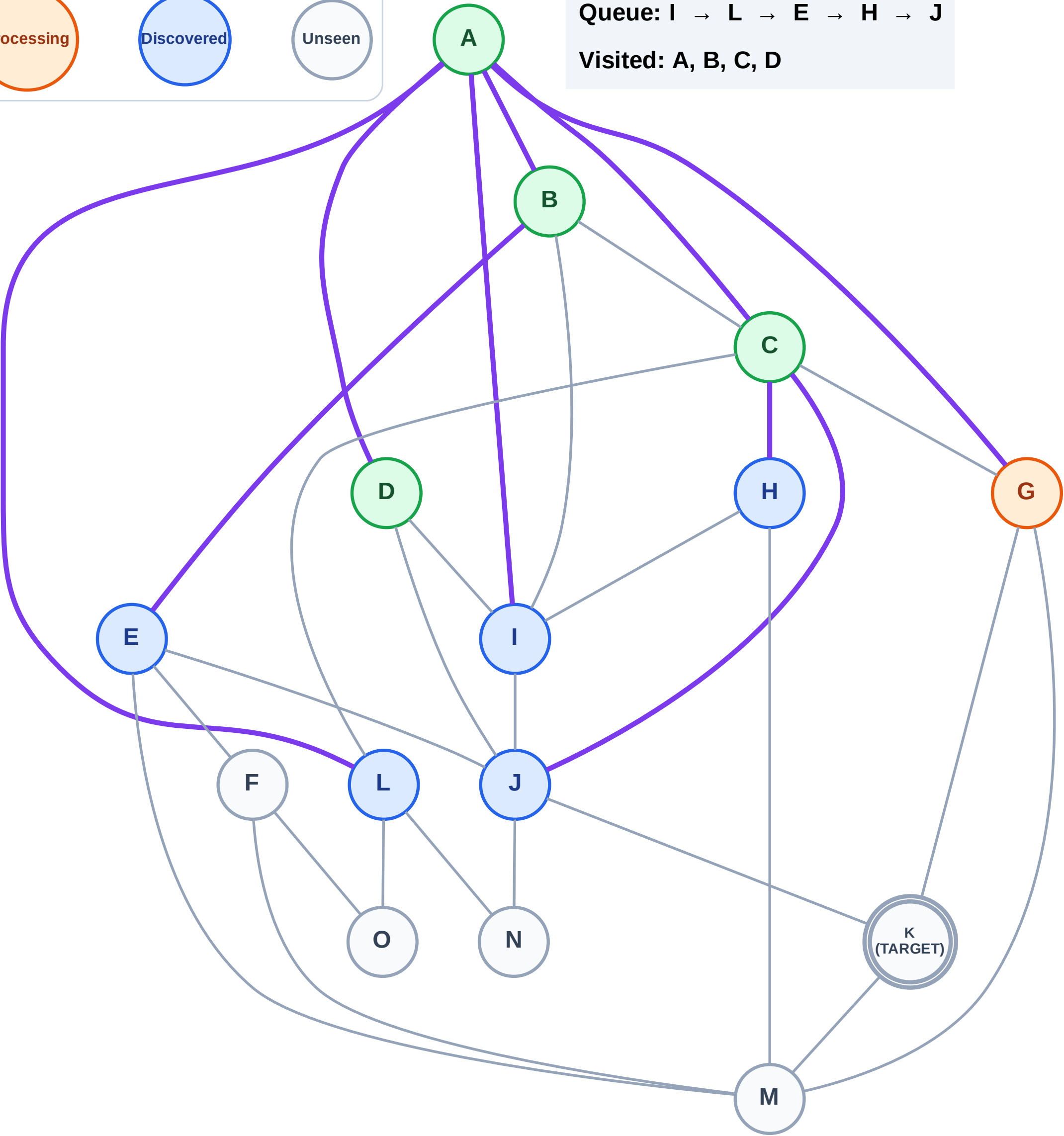
Processing

Discovered

Unseen

Queue: I → L → E → H → J

Visited: A, B, C, D



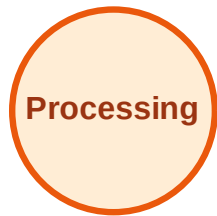
BFS Traversal

Step 11: Discover K, M from G

Legend



Complete



Processing



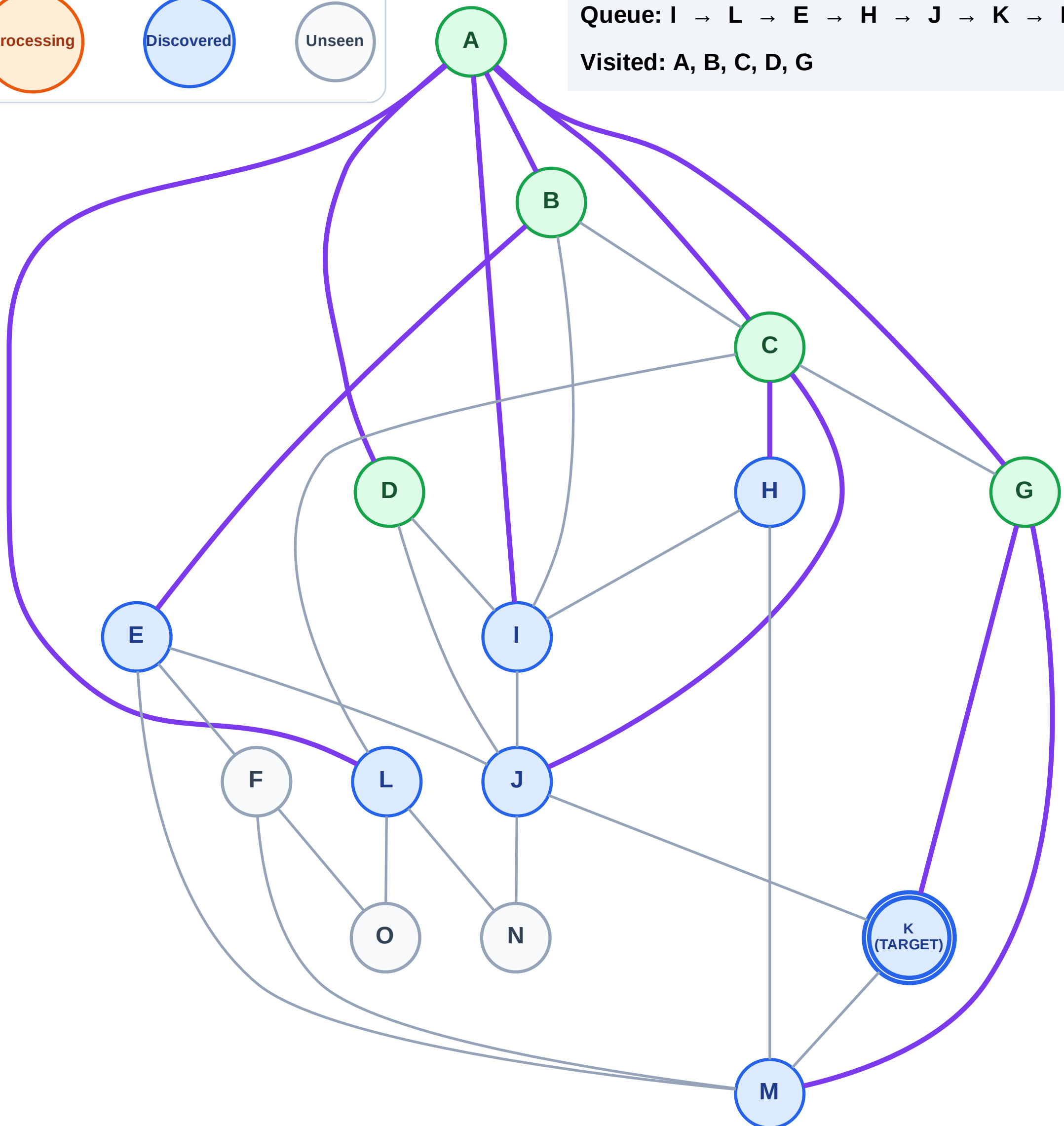
Discovered



Unseen

Queue: I → L → E → H → J → K → M

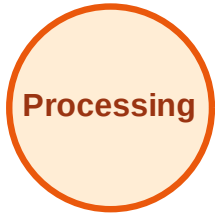
Visited: A, B, C, D, G



BFS Traversal

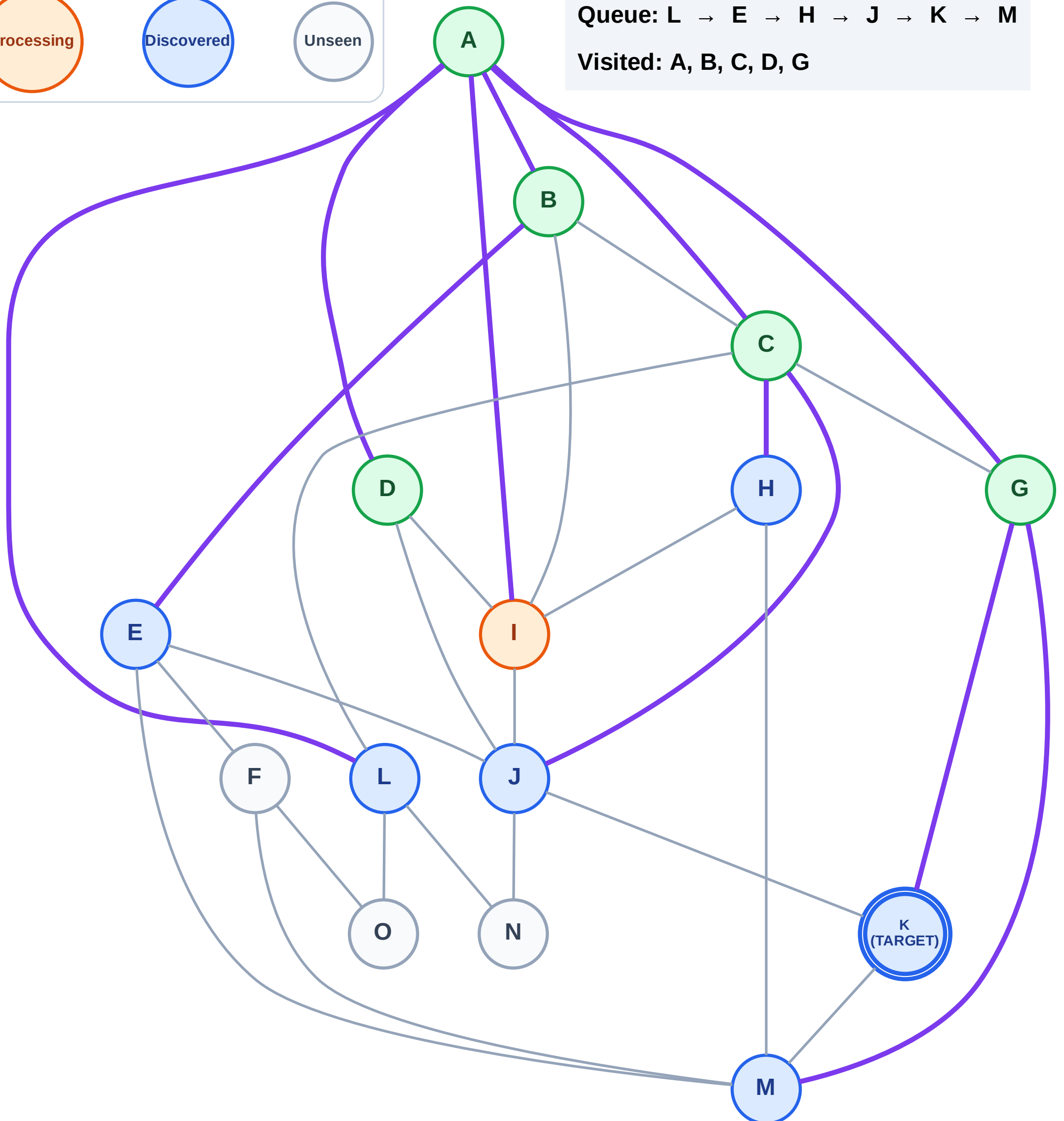
Step 12: Process node I

Legend



Queue: L → E → H → J → K → M

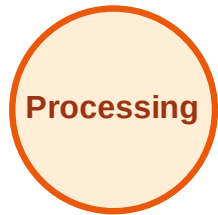
Visited: A, B, C, D, G



BFS Traversal

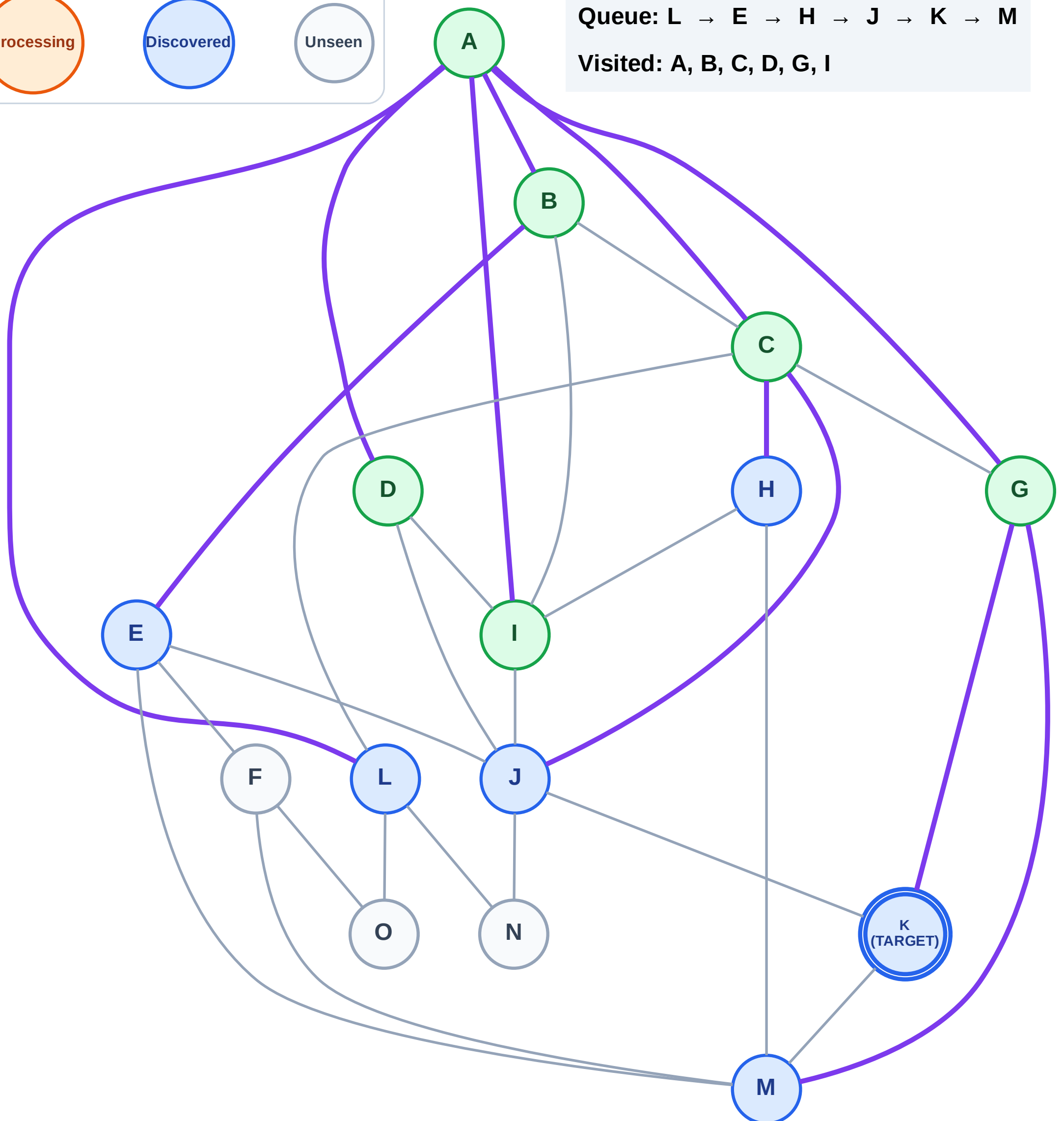
Step 13: No new neighbors from I

Legend



Queue: L → E → H → J → K → M

Visited: A, B, C, D, G, I



BFS Traversal

Step 14: Process node L

Legend

Complete

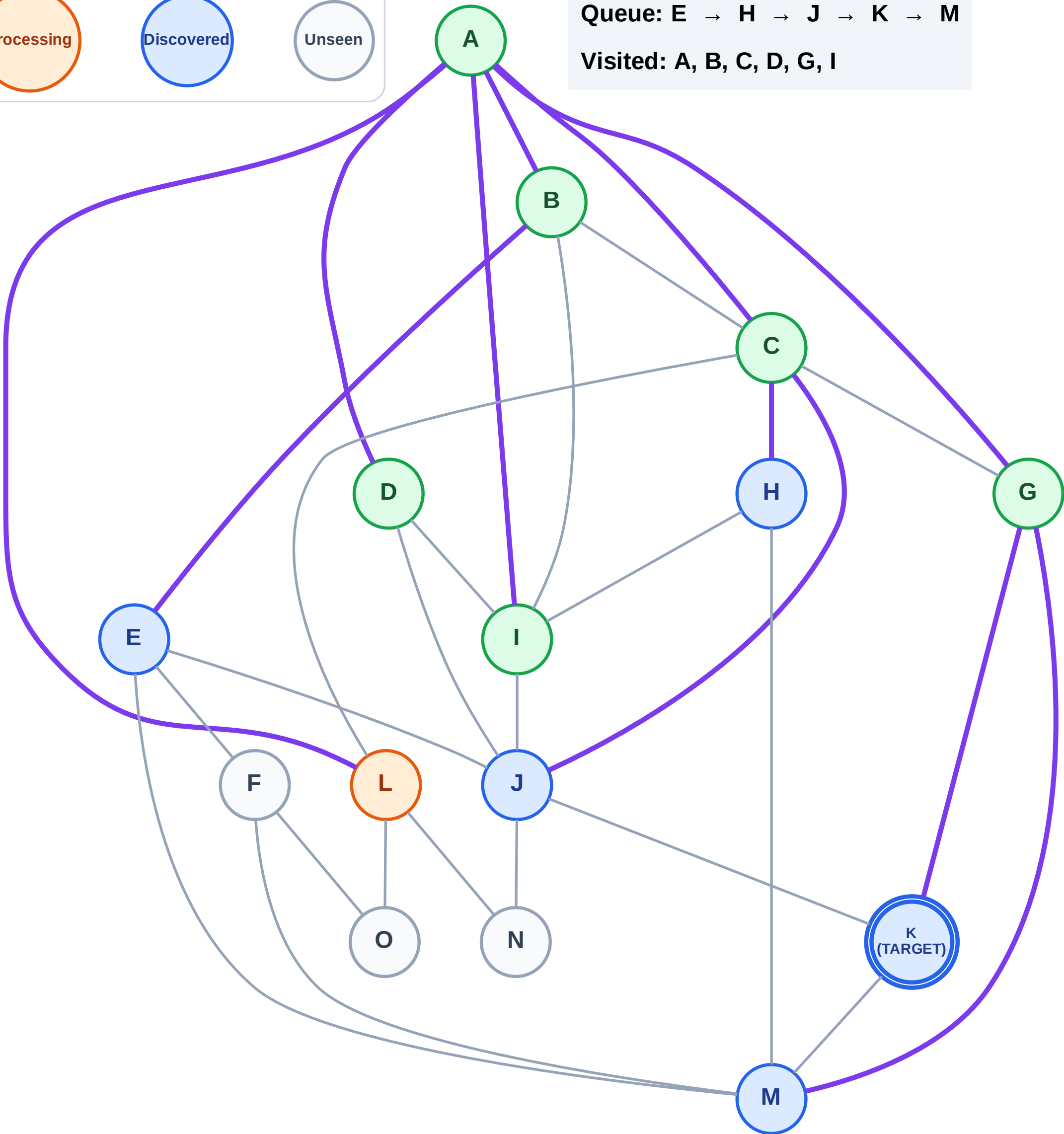
Processing

Discovered

Unseen

Queue: E → H → J → K → M

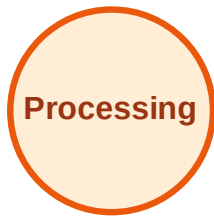
Visited: A, B, C, D, G, I



BFS Traversal

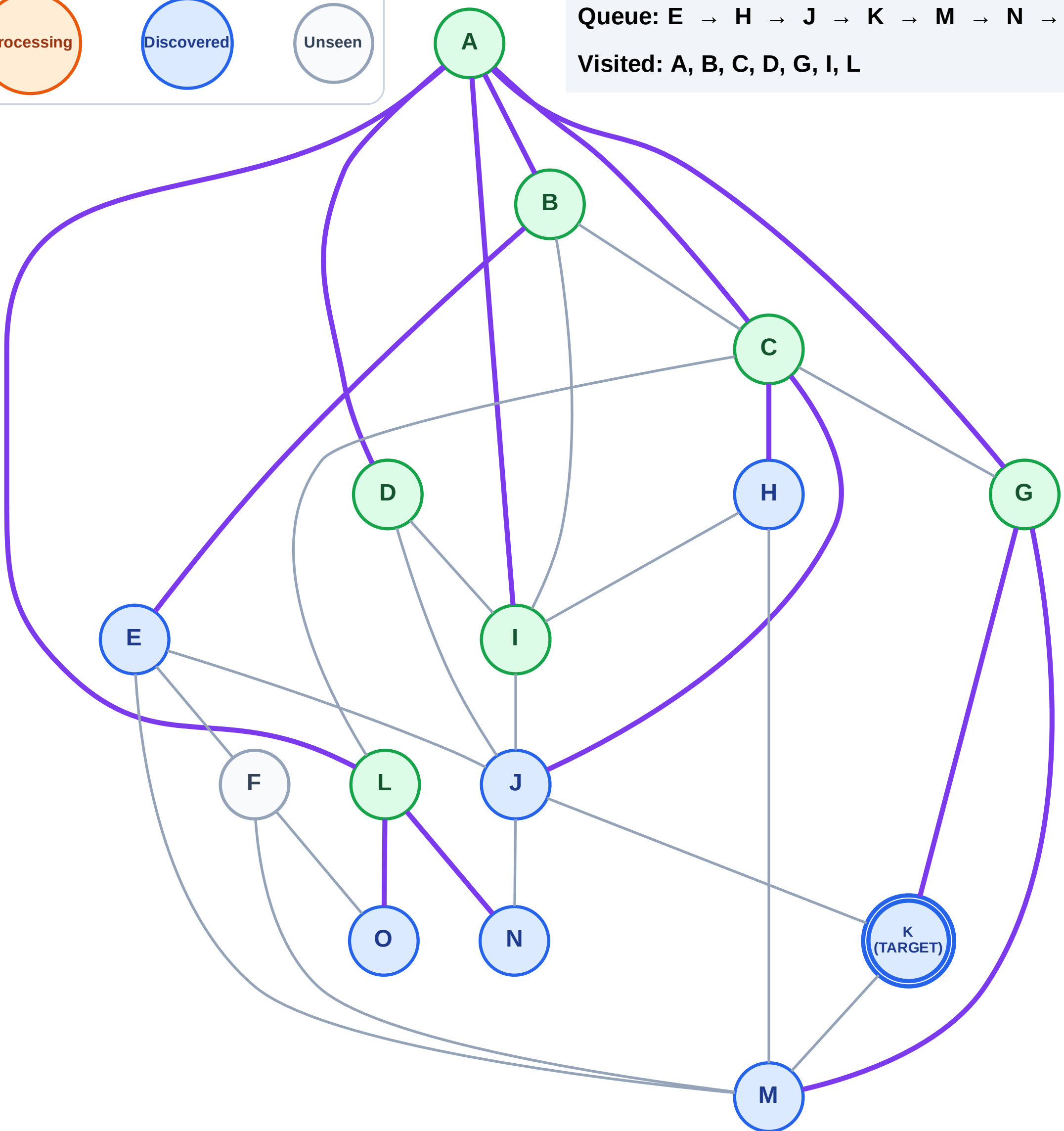
Step 15: Discover N, O from L

Legend



Queue: E → H → J → K → M → N → O

Visited: A, B, C, D, G, I, L



BFS Traversal

Step 16: Process node E

Legend

Complete

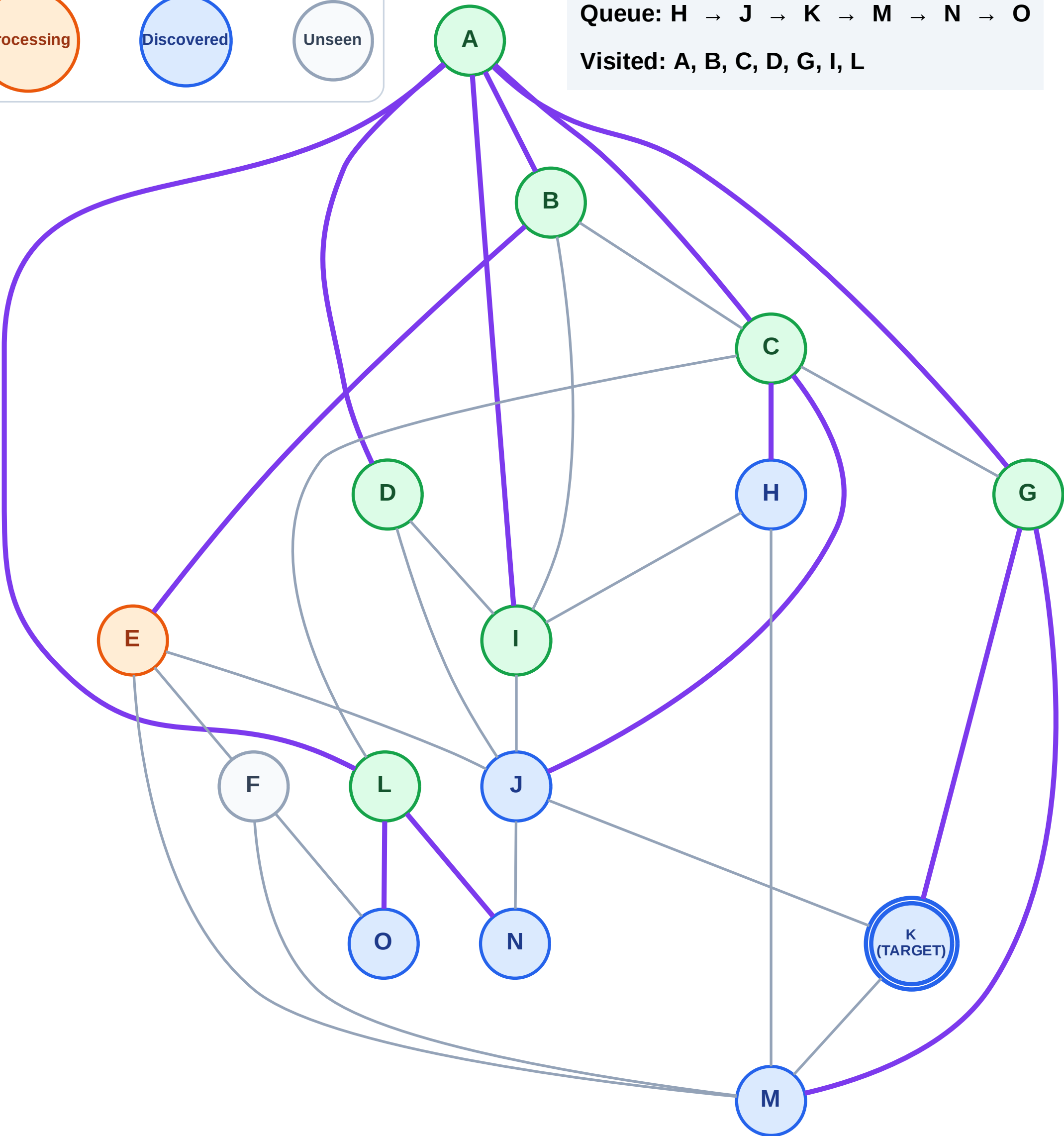
Processing

Discovered

Unseen

Queue: H → J → K → M → N → O

Visited: A, B, C, D, G, I, L

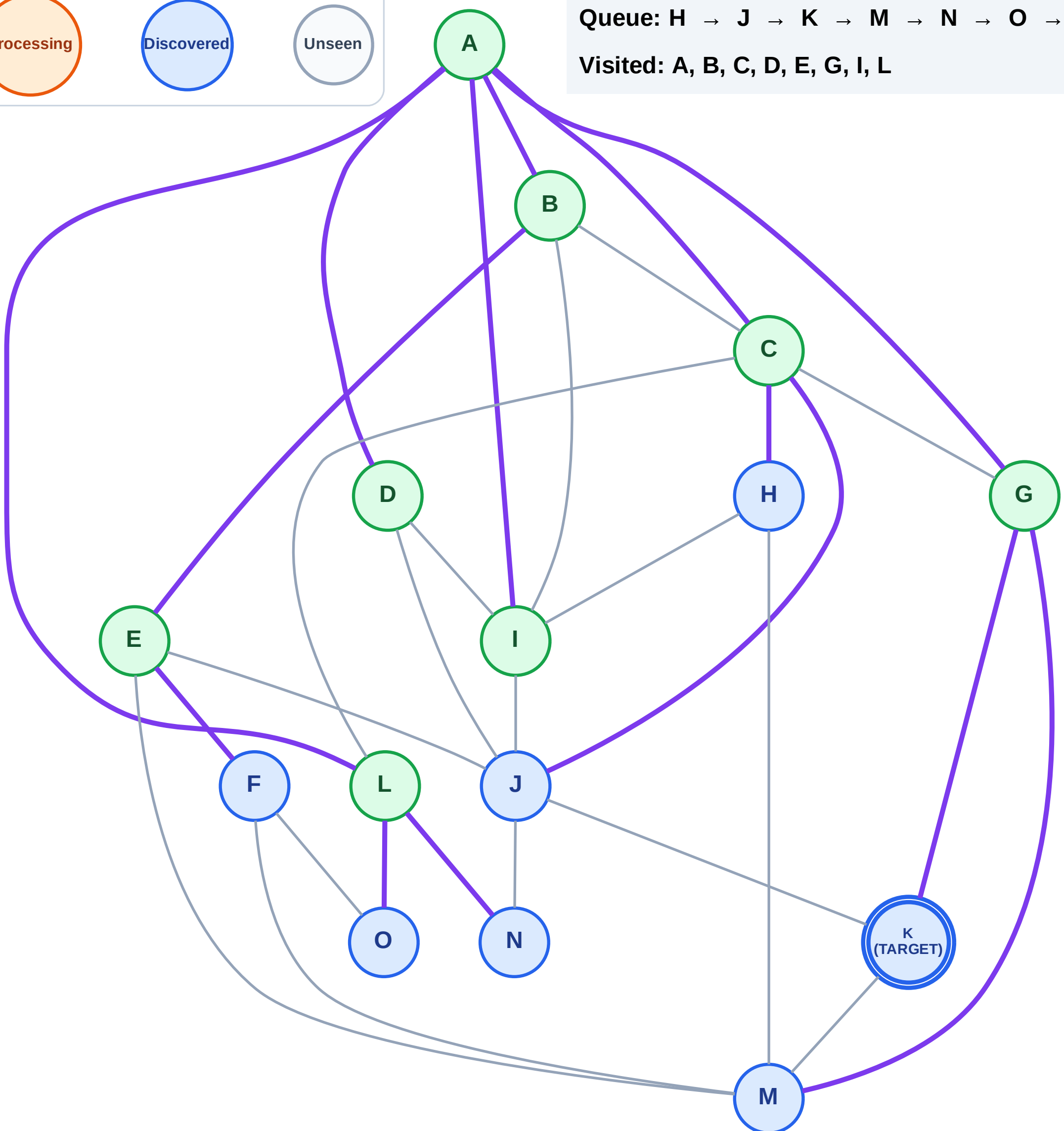


Step 17: Discover F from E

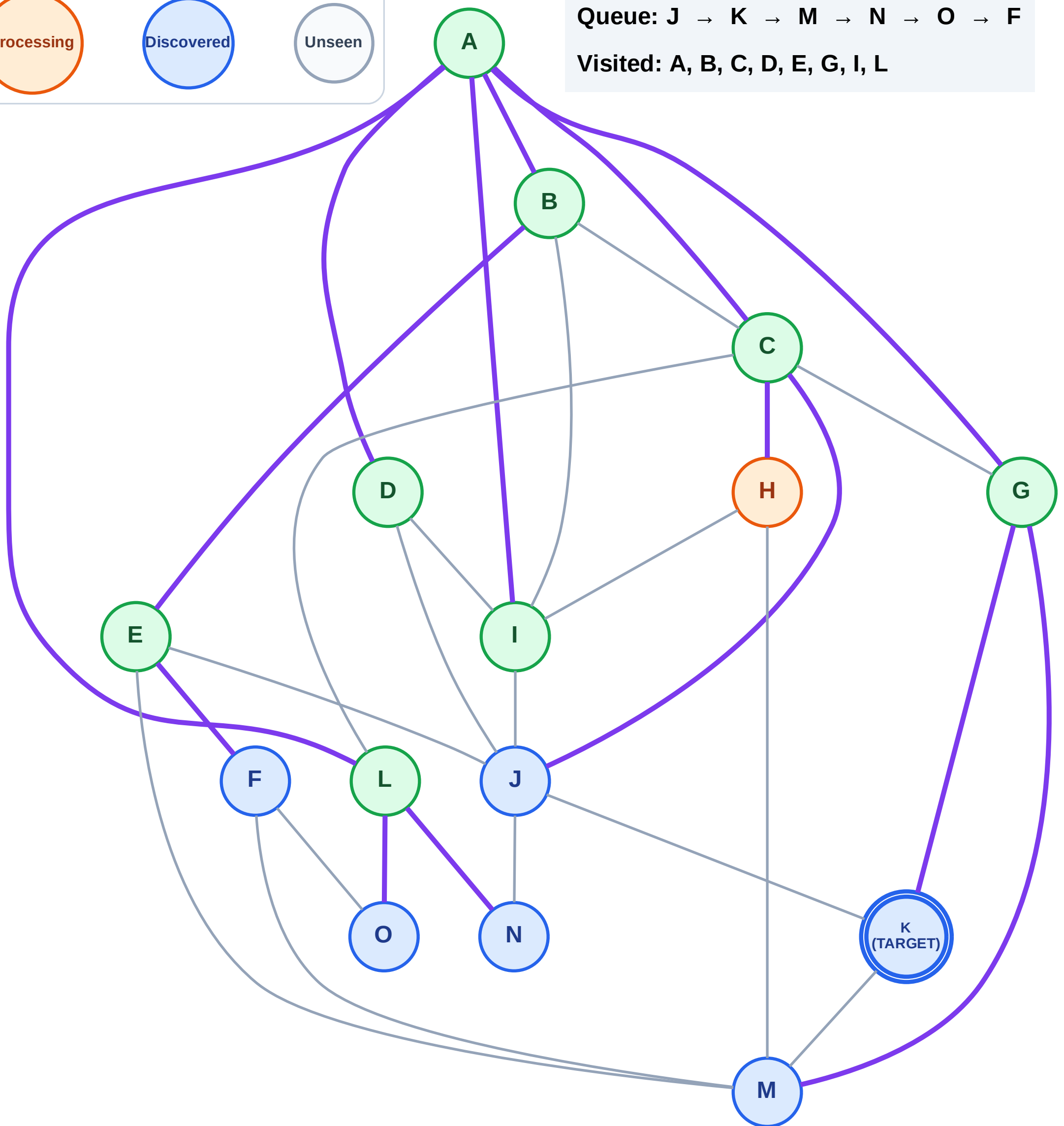
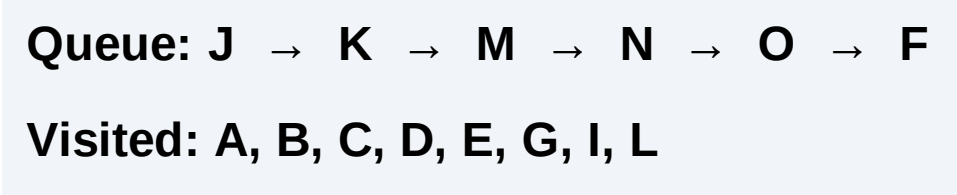
Queue: H → J → K → M → N → O → F

Legend

Unseen



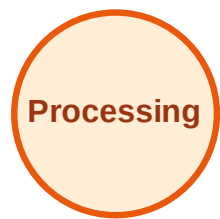
Step 18: Process node H



BFS Traversal

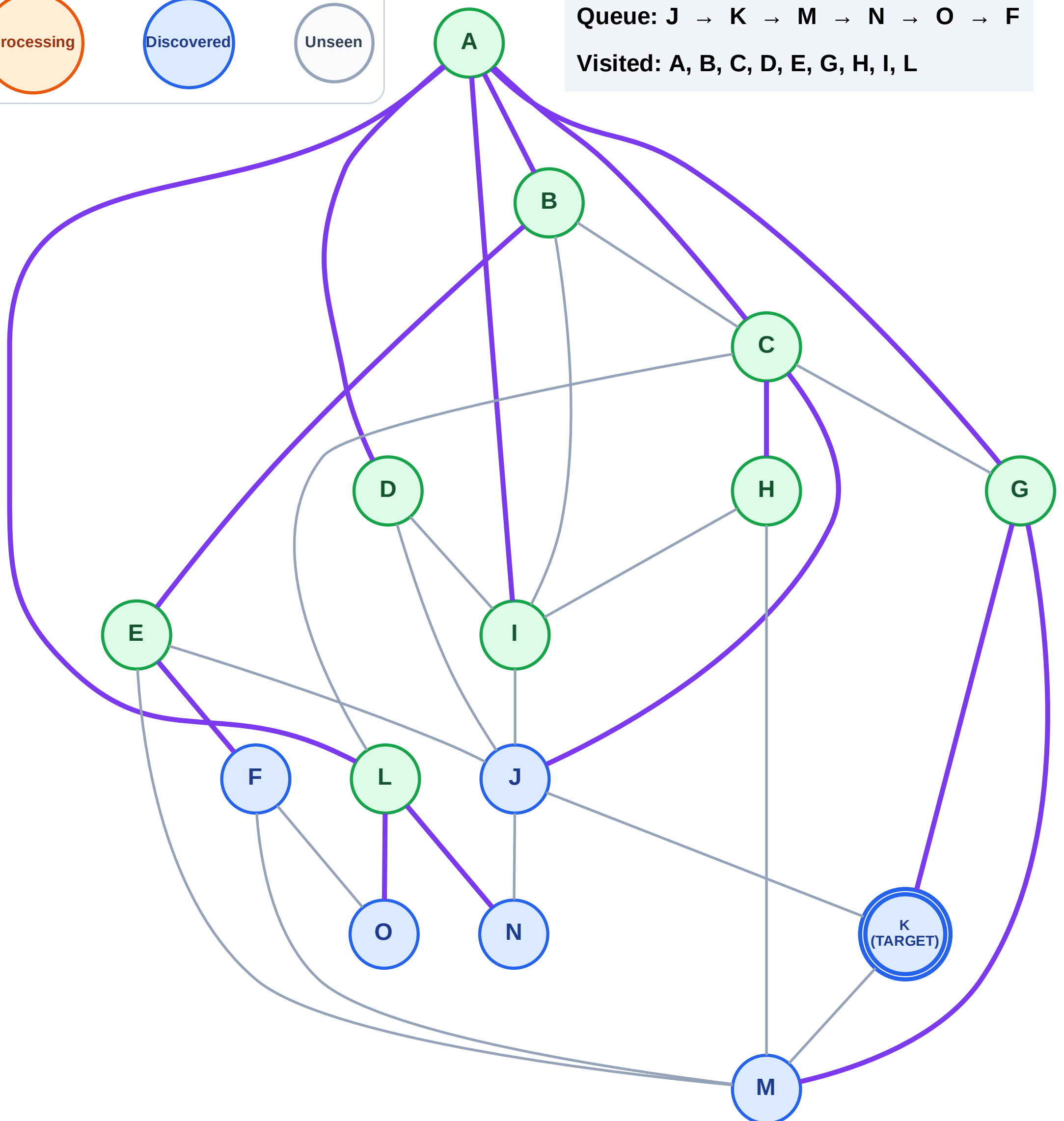
Step 19: No new neighbors from H

Legend



Queue: J → K → M → N → O → F

Visited: A, B, C, D, E, G, H, I, L



BFS Traversal

Step 20: Process node J

Legend

Complete

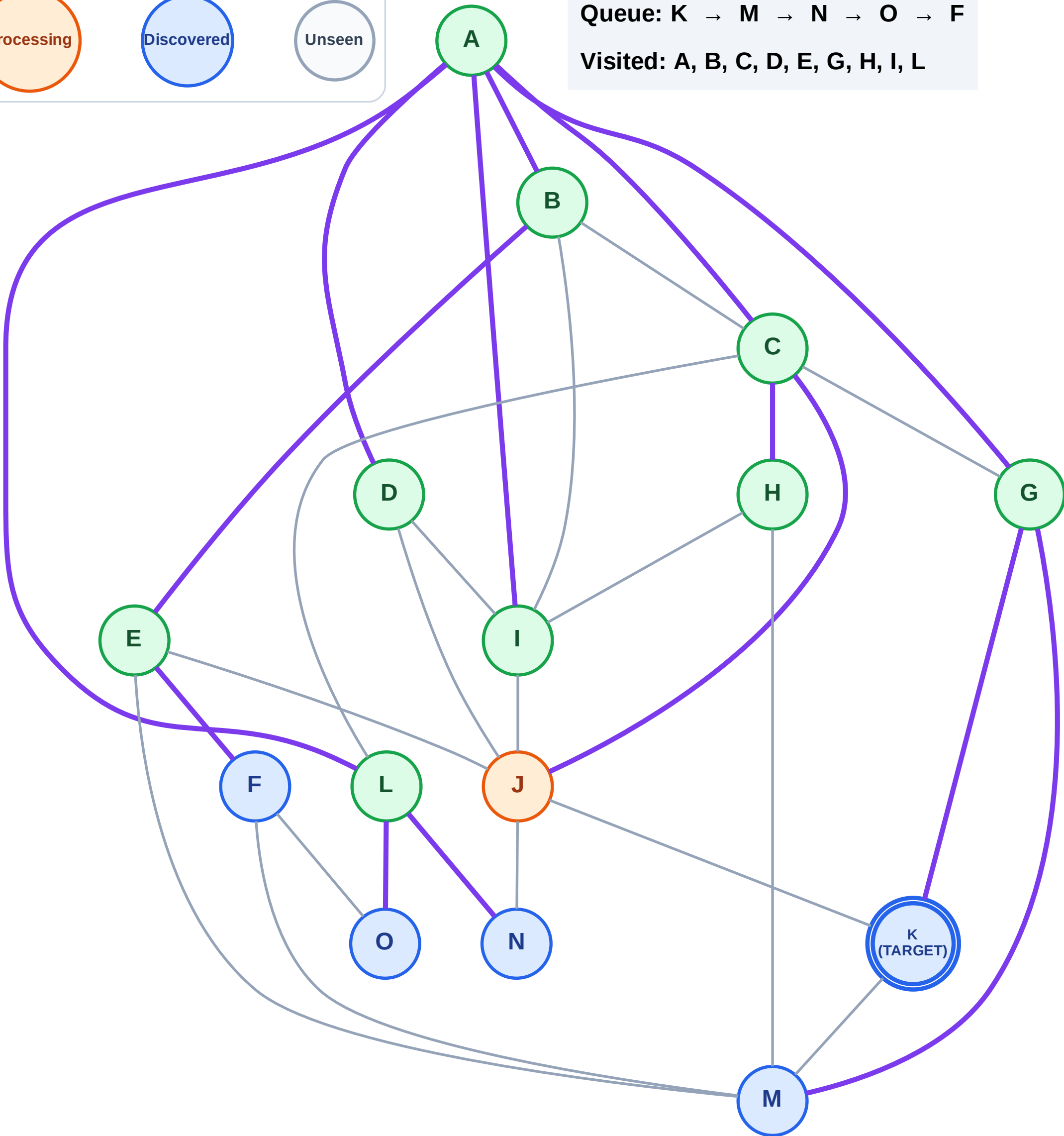
Processing

Discovered

Unseen

Queue: K → M → N → O → F

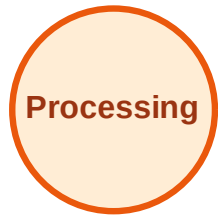
Visited: A, B, C, D, E, G, H, I, L



BFS Traversal

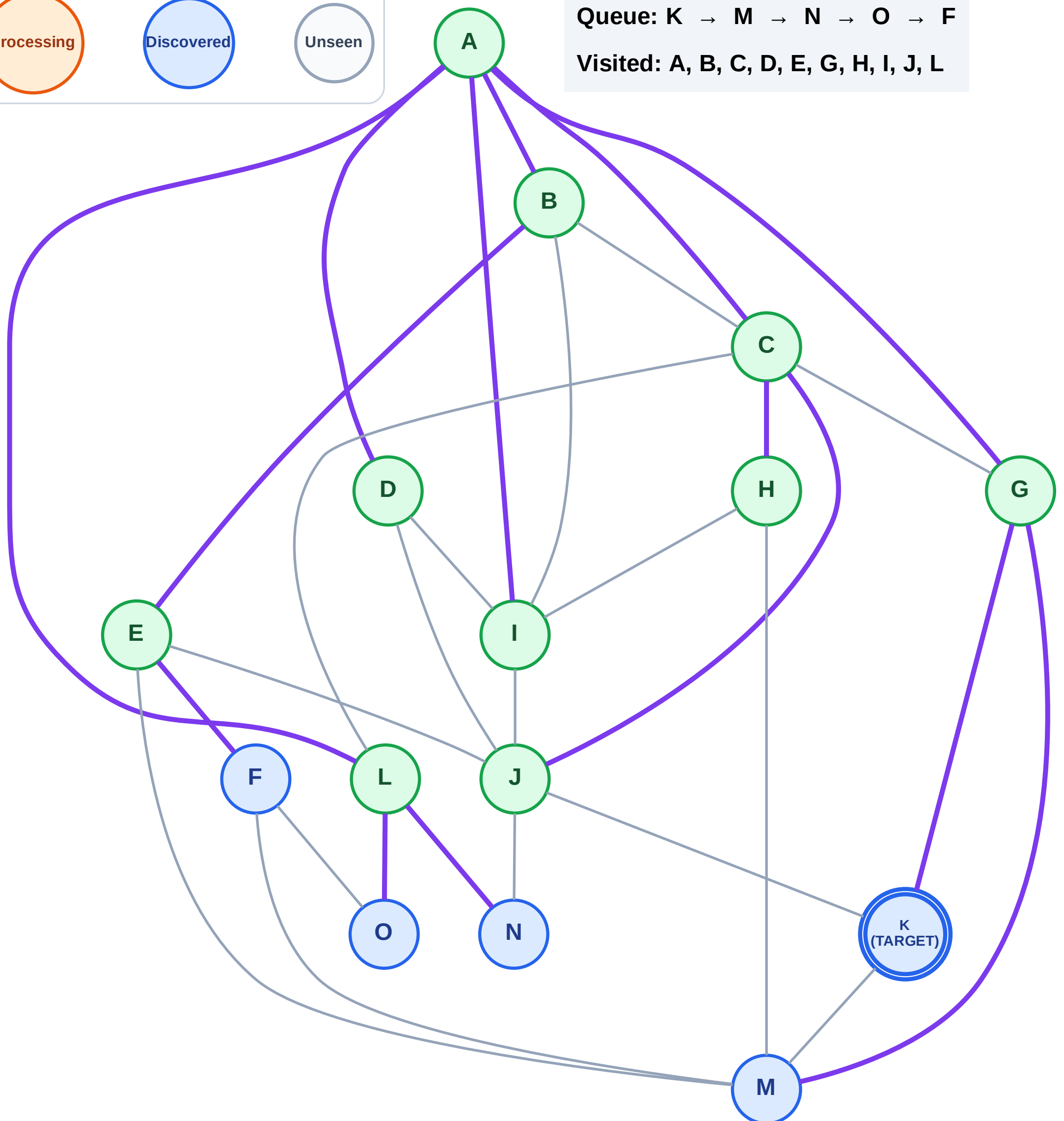
Step 21: No new neighbors from J

Legend



Queue: K → M → N → O → F

Visited: A, B, C, D, E, G, H, I, J, L



BFS Traversal

Step 22: Process node K

Legend

Complete

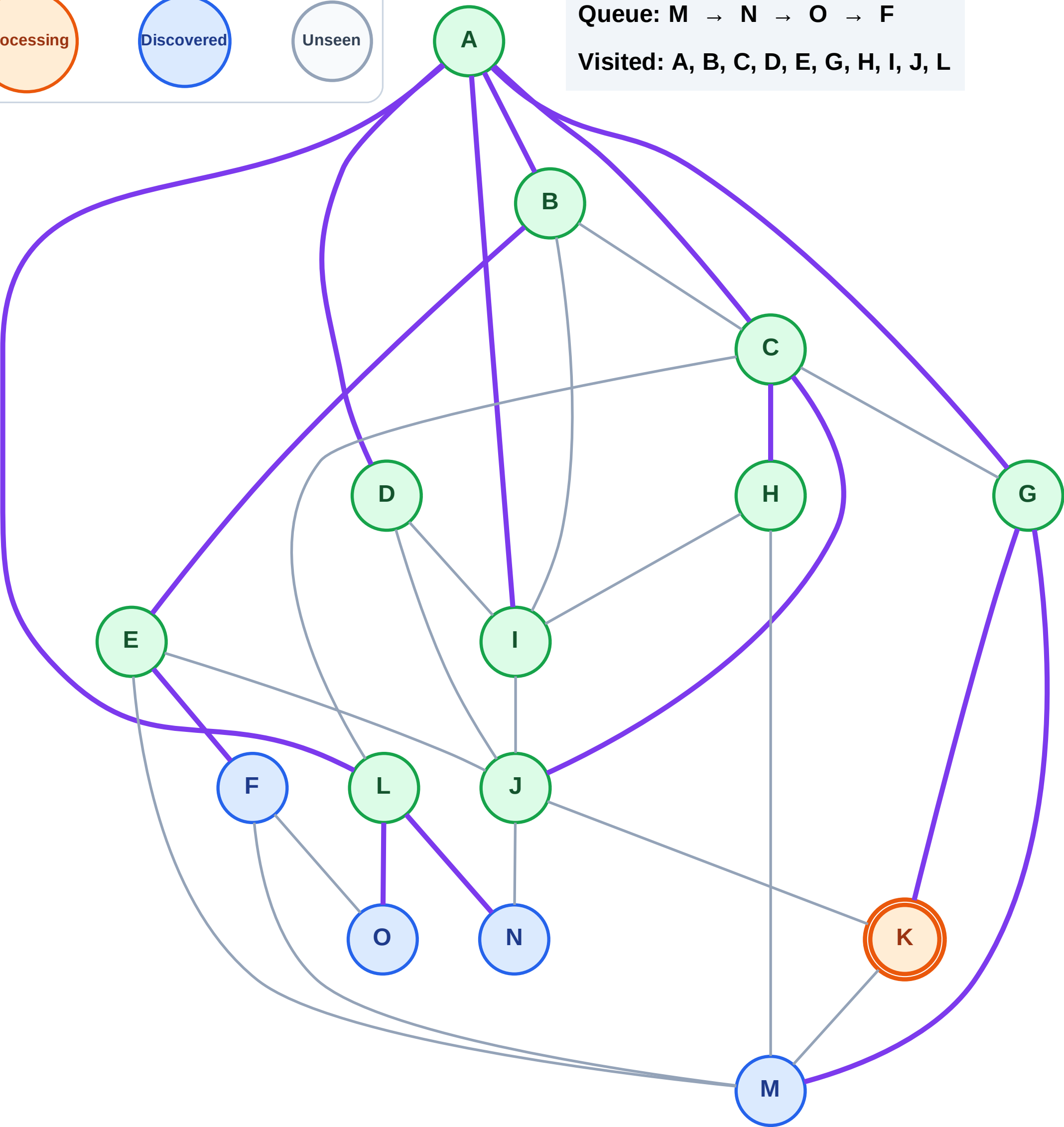
Processing

Discovered

Unseen

Queue: M → N → O → F

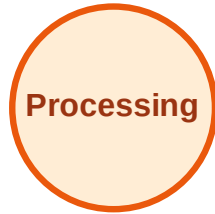
Visited: A, B, C, D, E, G, H, I, J, L



BFS Traversal

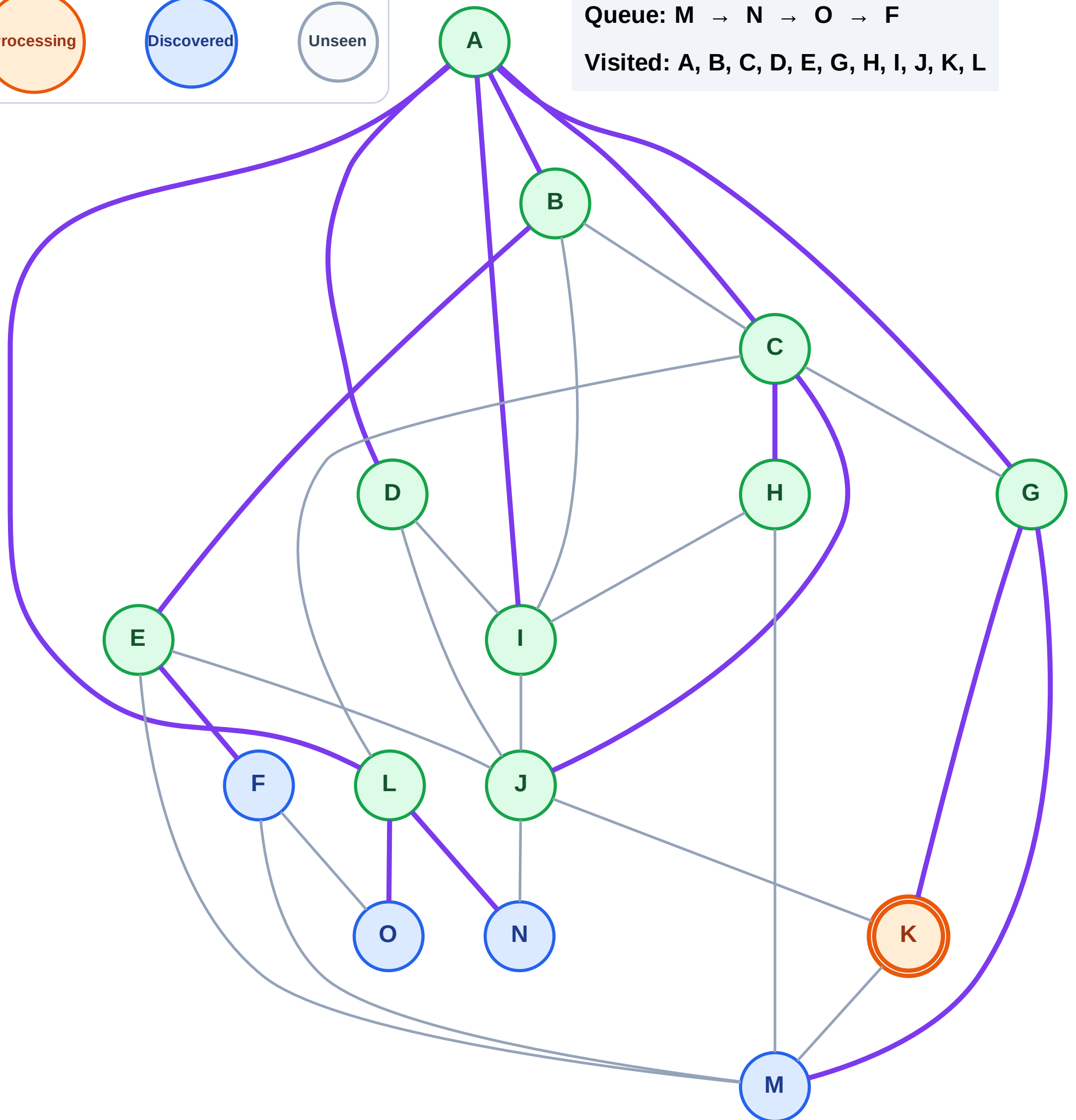
Step 23: TARGET NODE K REACHED! Path (2 hops): A → G → K

Legend



Queue: M → N → O → F

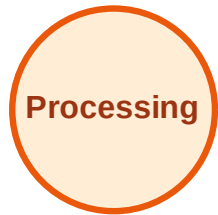
Visited: A, B, C, D, E, G, H, I, J, K, L



DFS Traversal

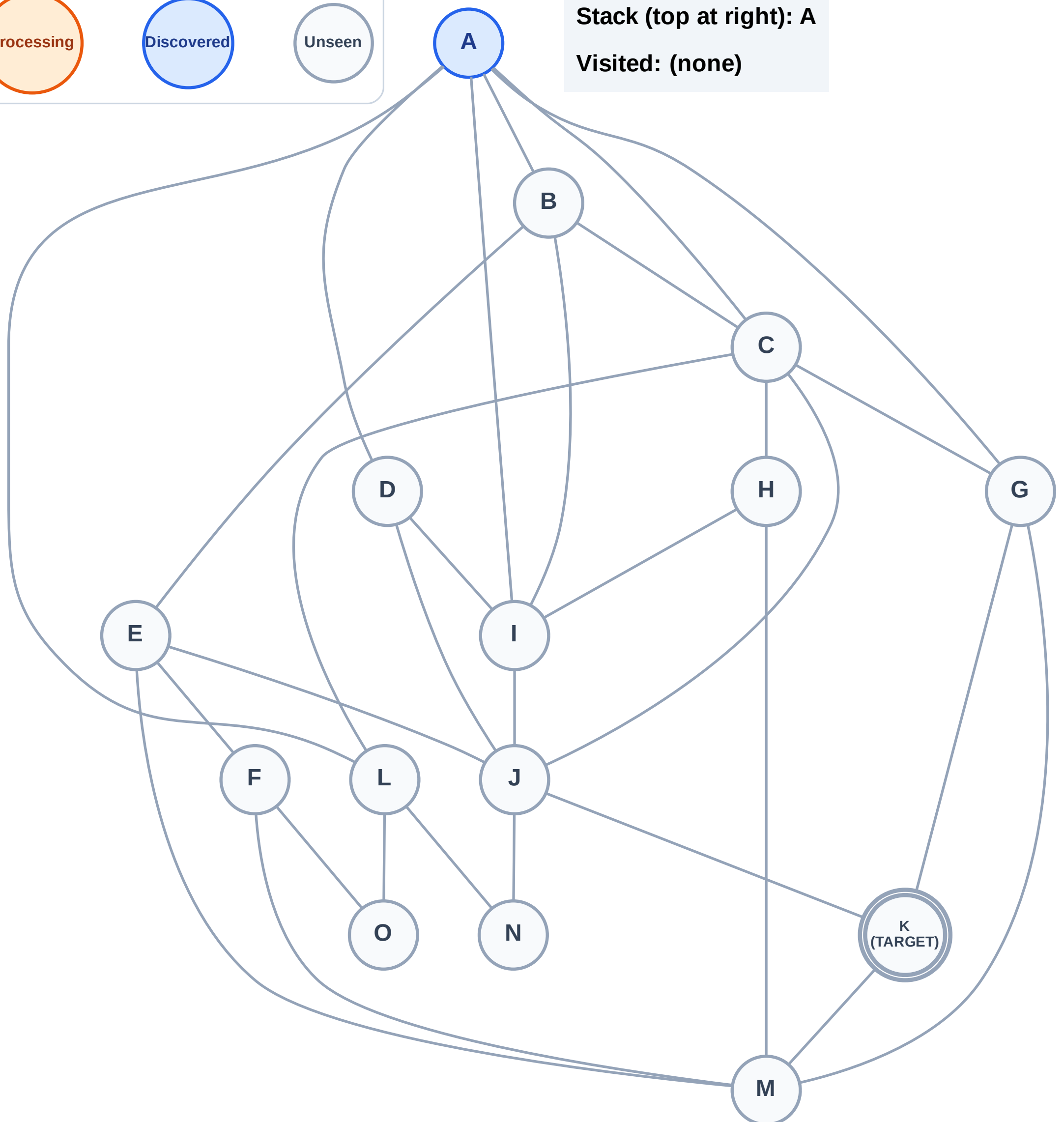
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



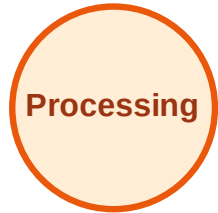
DFS Traversal

Step 2: Process node A

Legend



Complete



Processing



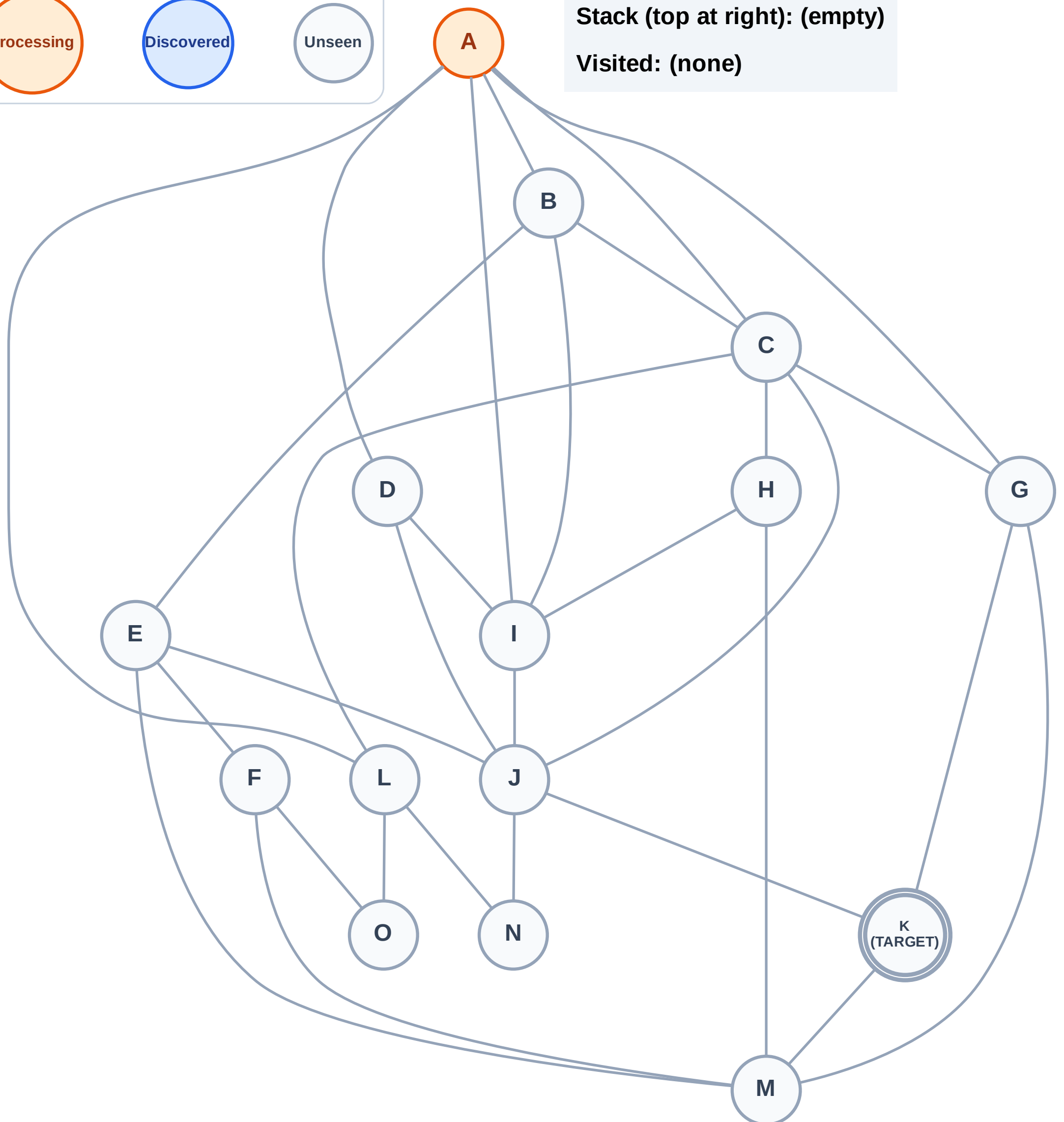
Discovered



Unseen

Stack (top at right): (empty)

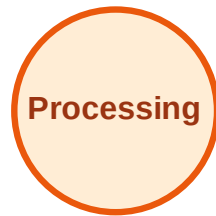
Visited: (none)



DFS Traversal

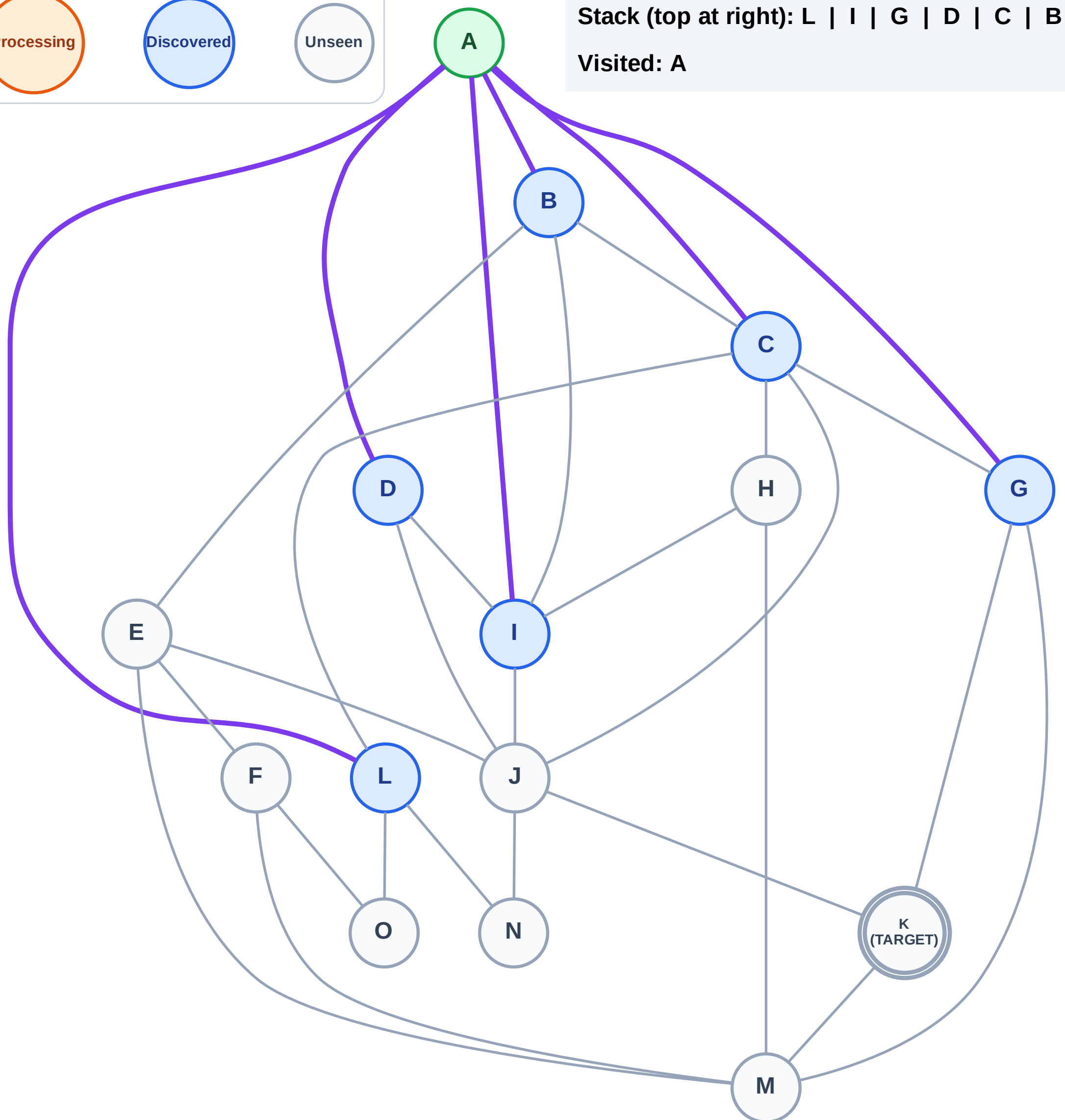
Step 3: Discover L, I, G, D, C, B from A

Legend



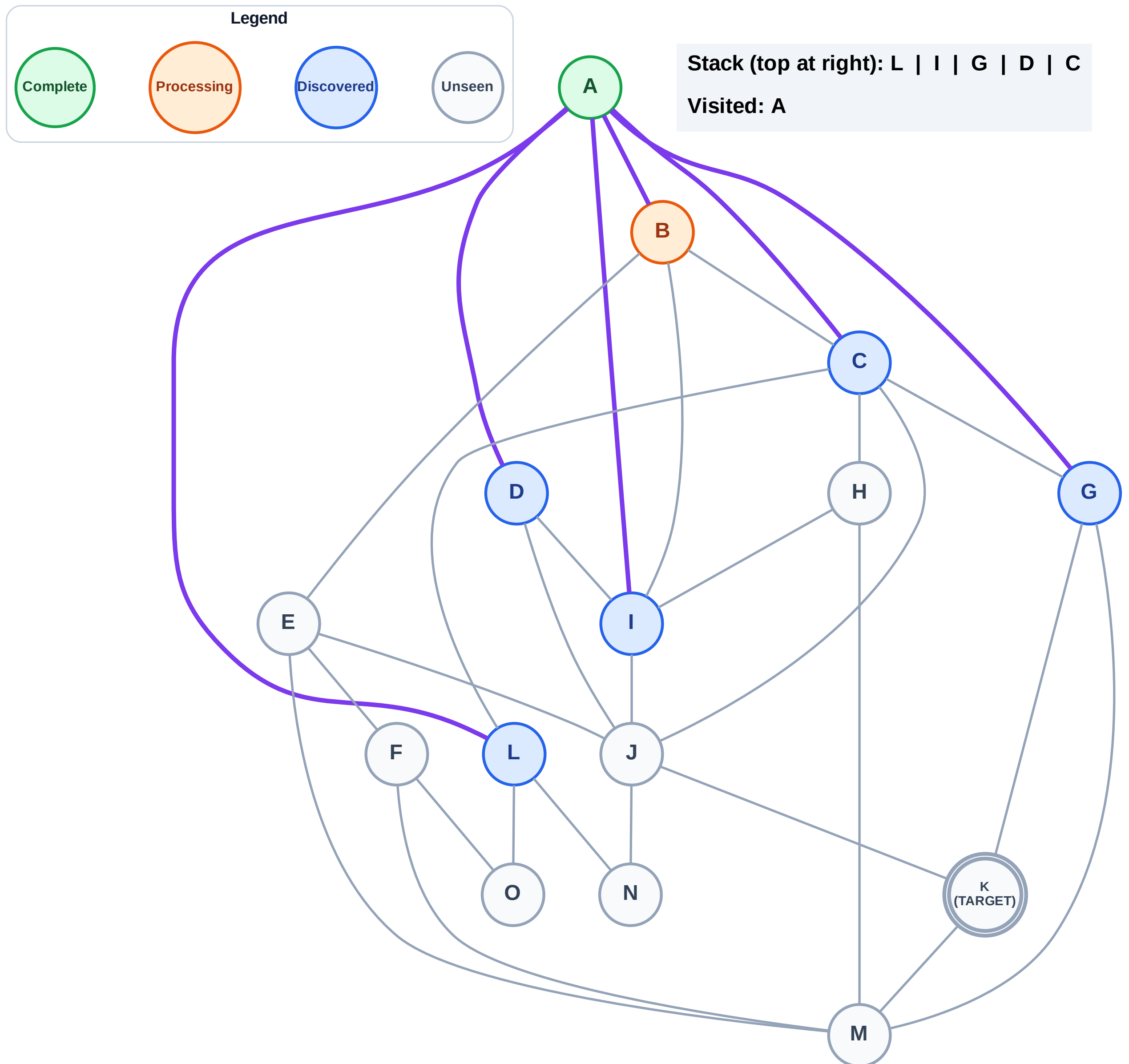
Stack (top at right): L | I | G | D | C | B

Visited: A



Step 4: Process node B

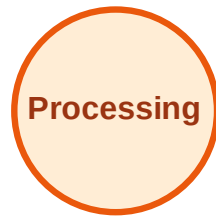
Step 4: Process node B



DFS Traversal

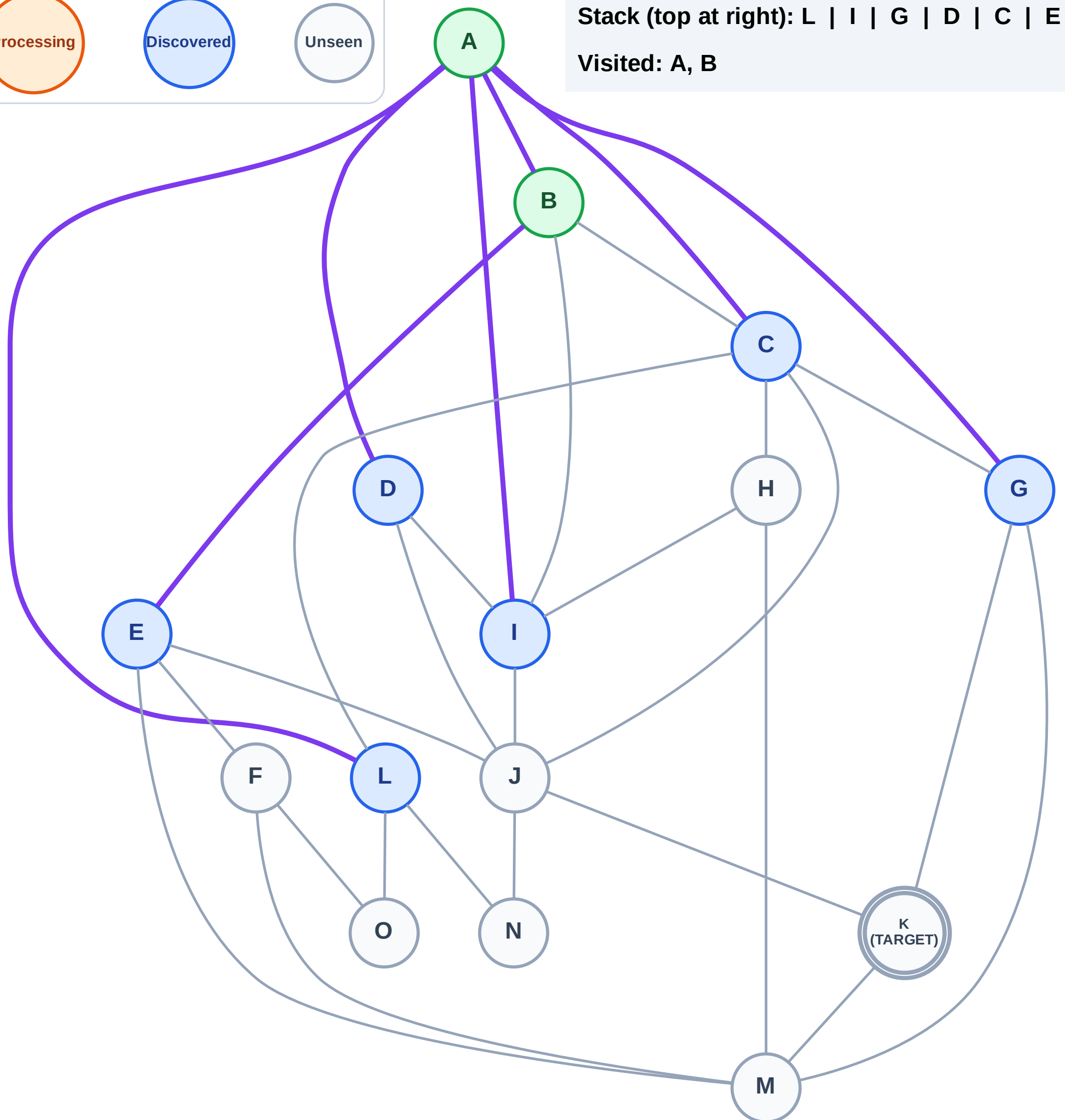
Step 5: Discover E from B

Legend



Stack (top at right): L | I | G | D | C | E

Visited: A, B



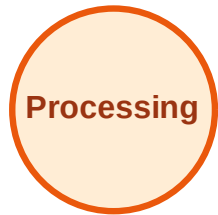
DFS Traversal

Step 6: Process node E

Legend



Complete



Processing



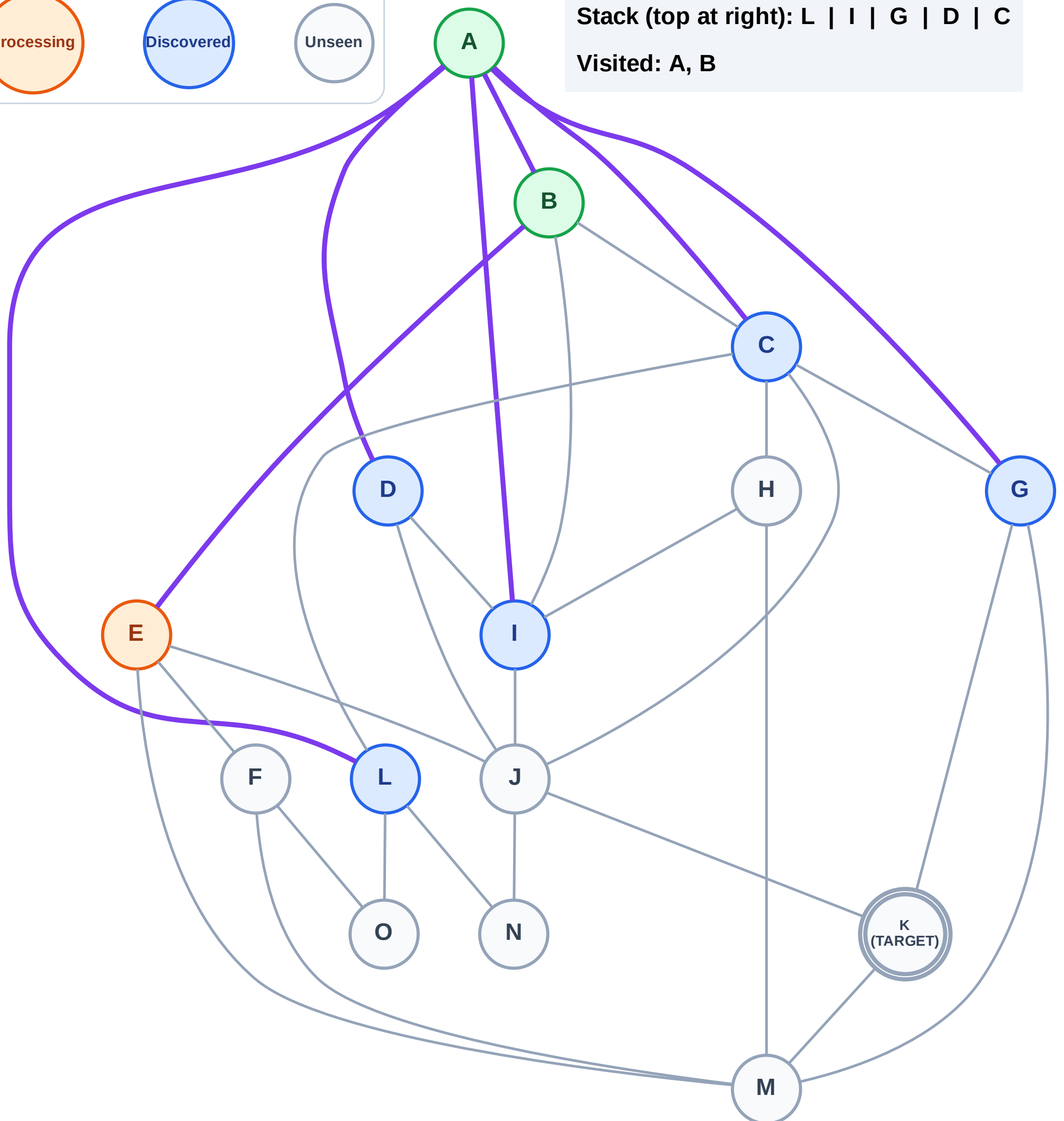
Discovered



Unseen

Stack (top at right): L | I | G | D | C

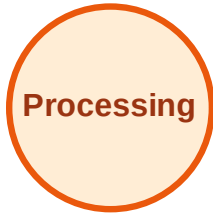
Visited: A, B



DFS Traversal

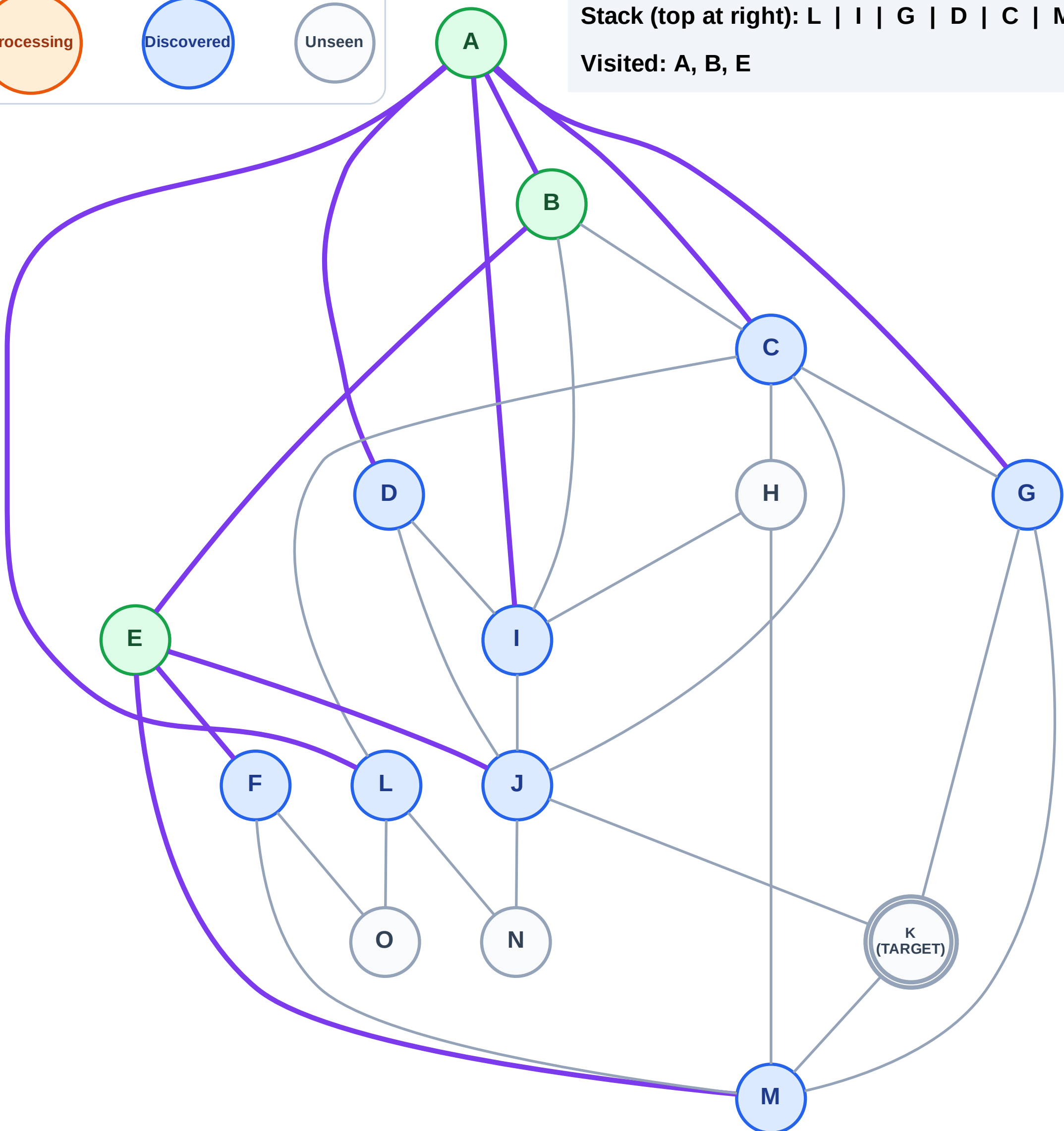
Step 7: Discover M, J, F from E

Legend



Stack (top at right): L | I | G | D | C | M | J | F

Visited: A, B, E



DFS Traversal

Step 8: Process node F

Legend

Complete

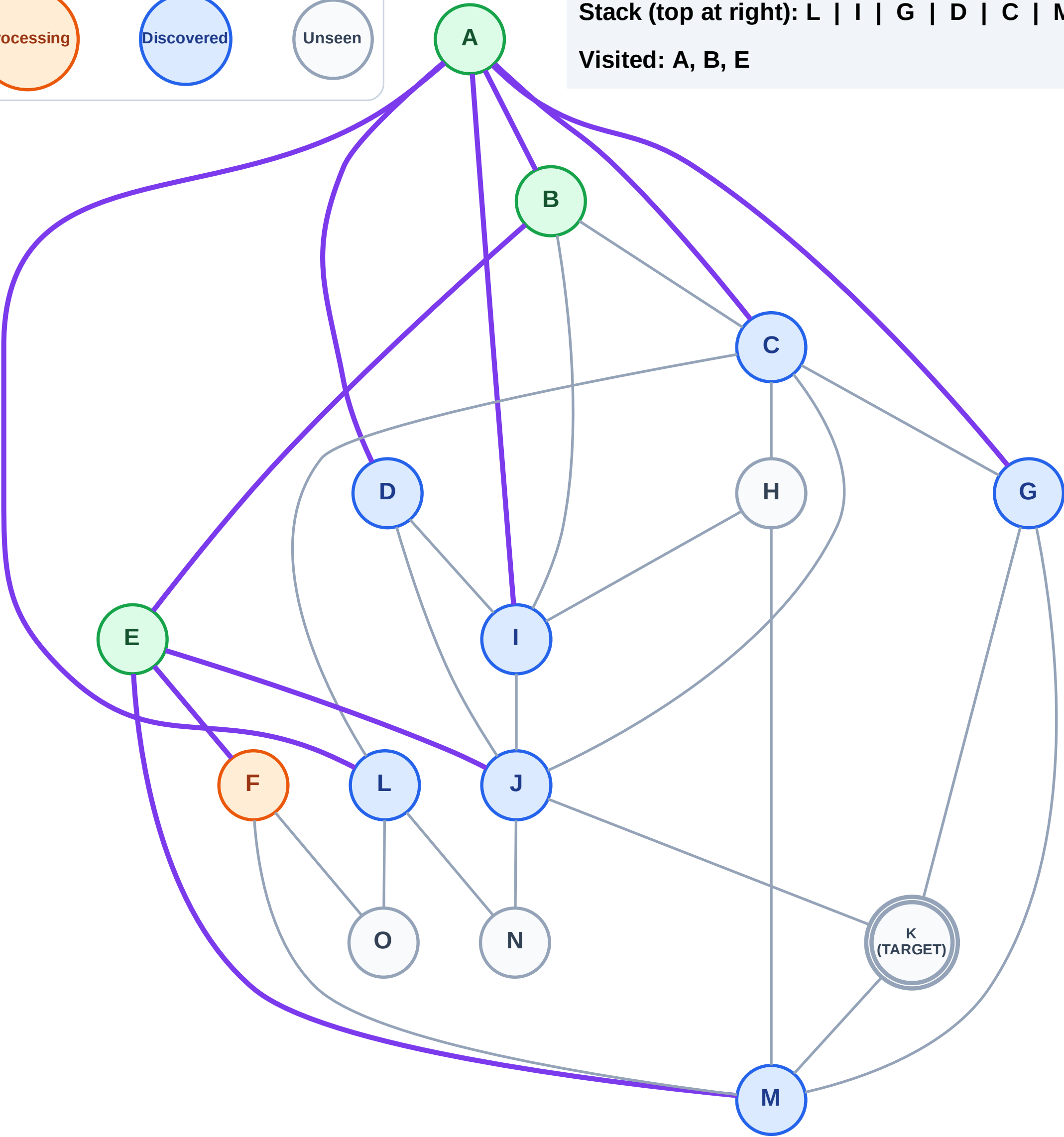
Processing

Discovered

Unseen

Stack (top at right): L | I | G | D | C | M | J

Visited: A, B, E



DFS Traversal

Step 9: Discover O from F

Legend

Complete

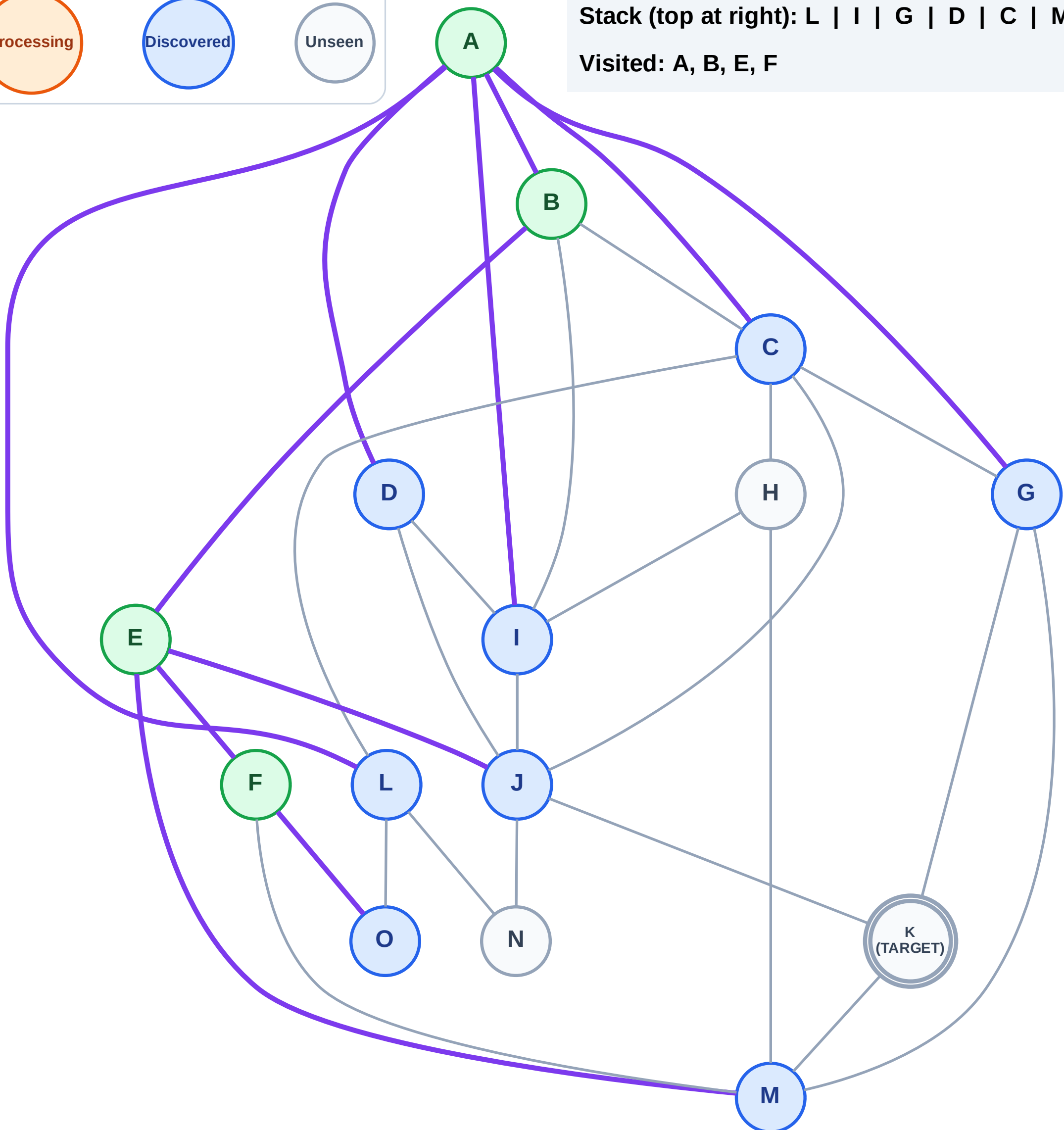
Processing

Discovered

Unseen

Stack (top at right): L | I | G | D | C | M | J | O

Visited: A, B, E, F



DFS Traversal

Step 10: Process node O

Legend



Complete



Processing



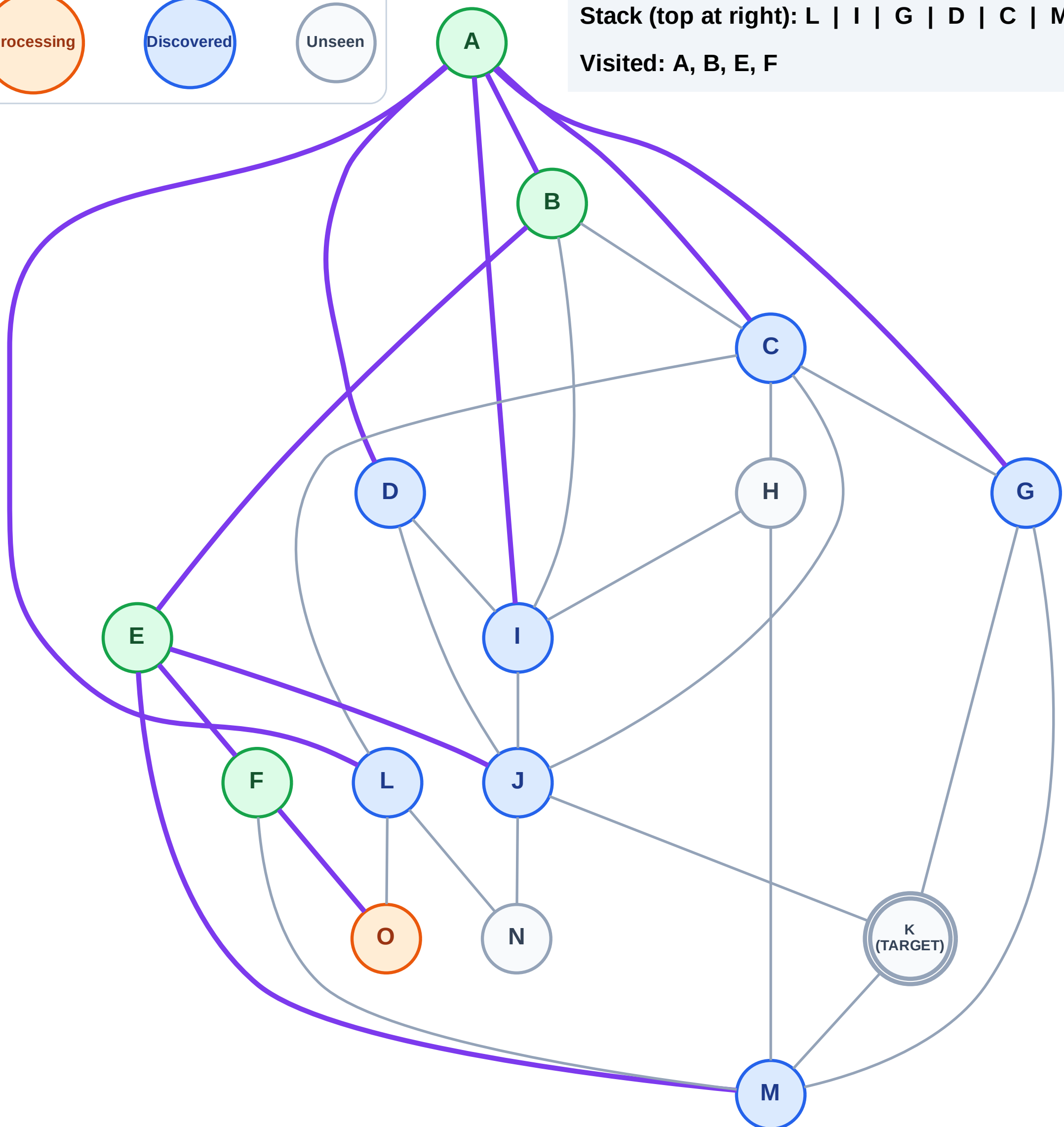
Discovered



Unseen

Stack (top at right): L | I | G | D | C | M | J

Visited: A, B, E, F



DFS Traversal

Step 11: No new neighbors from O

Legend



Complete



Processing



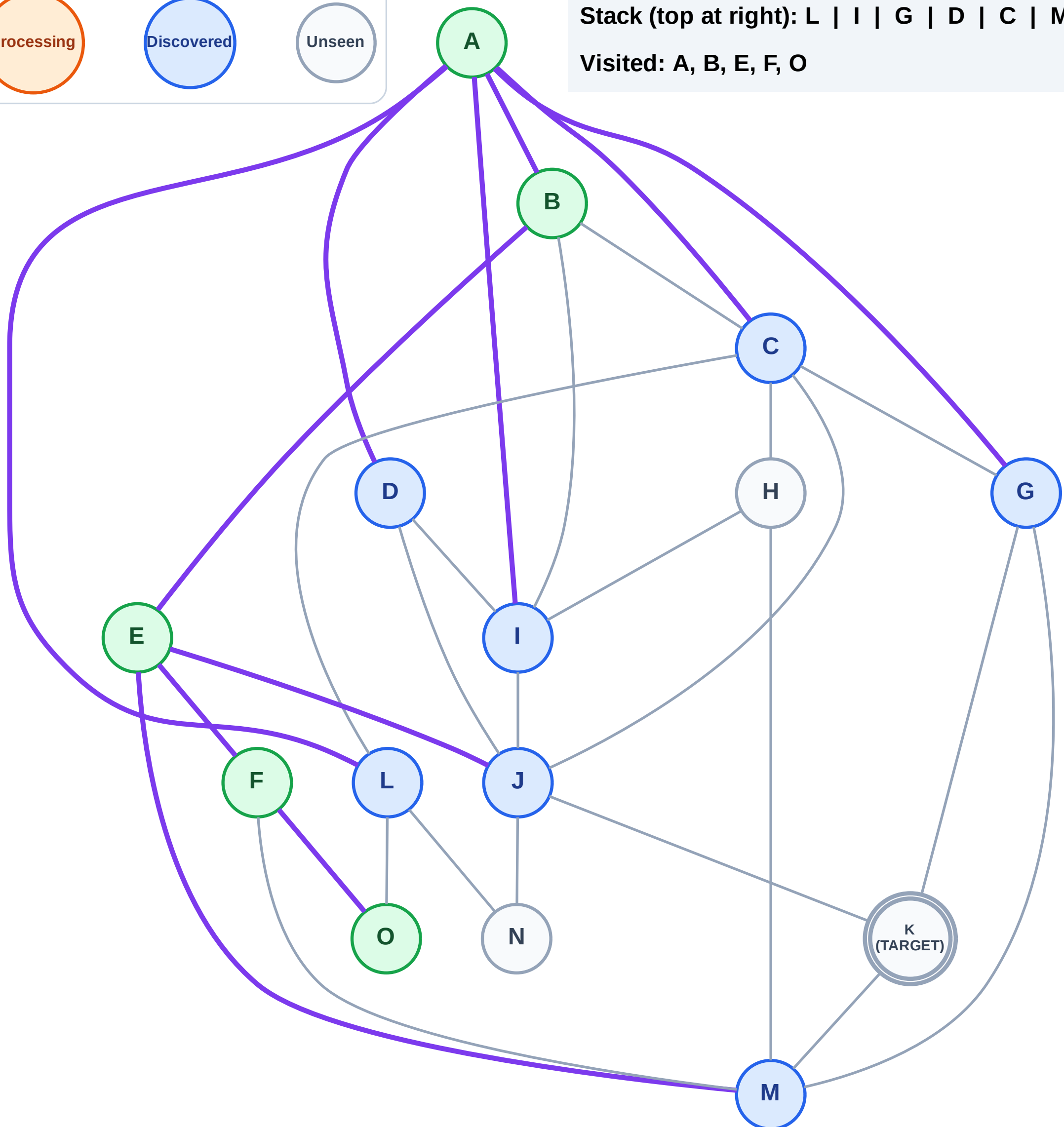
Discovered



Unseen

Stack (top at right): L | I | G | D | C | M | J

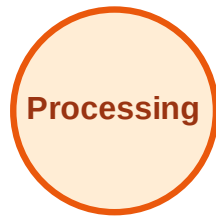
Visited: A, B, E, F, O



DFS Traversal

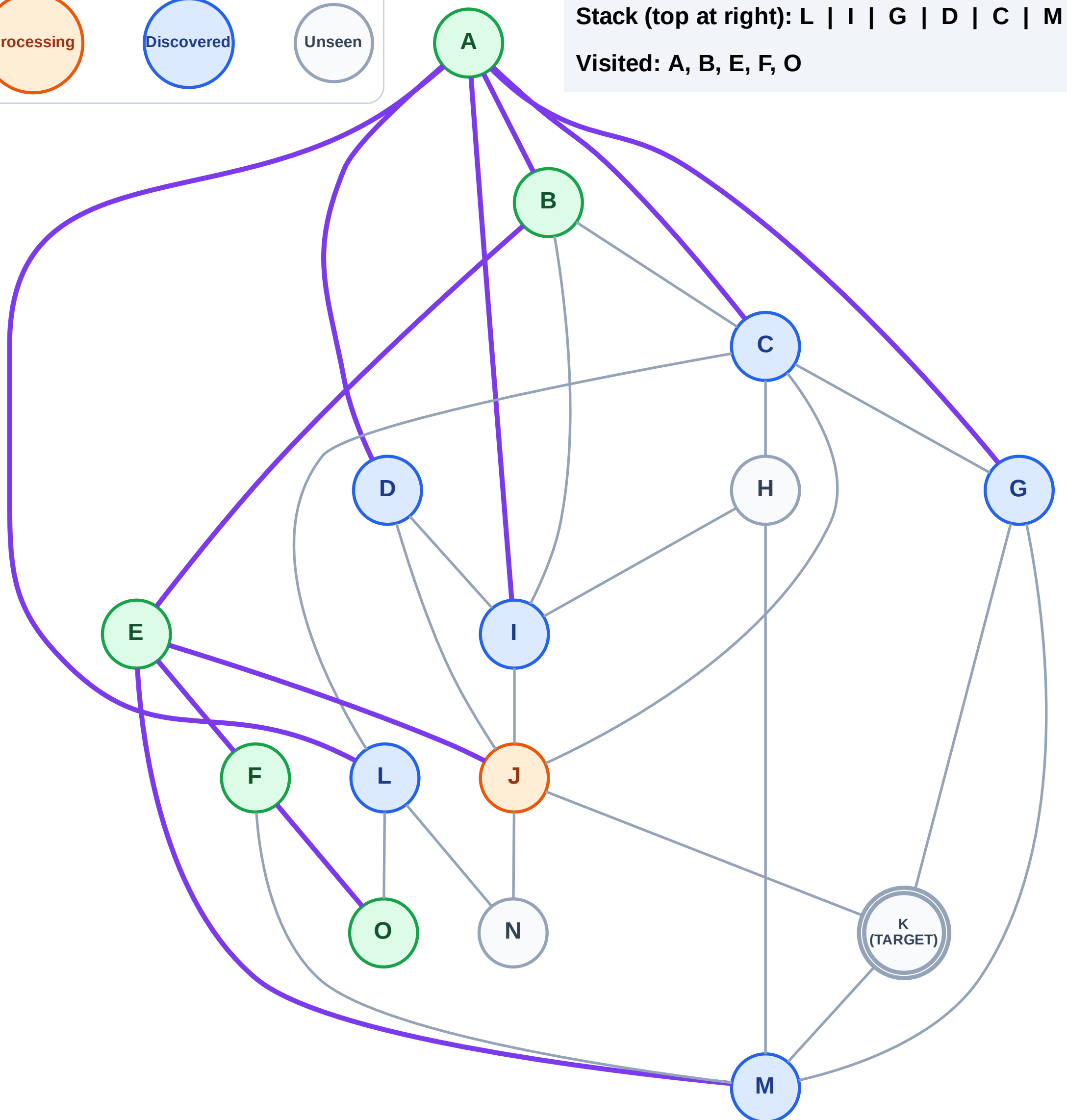
Step 12: Process node J

Legend



Stack (top at right): L | I | G | D | C | M

Visited: A, B, E, F, O



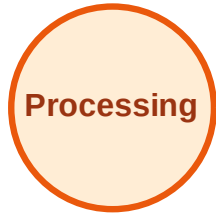
DFS Traversal

Step 13: Discover N, K from J

Legend



Complete



Processing



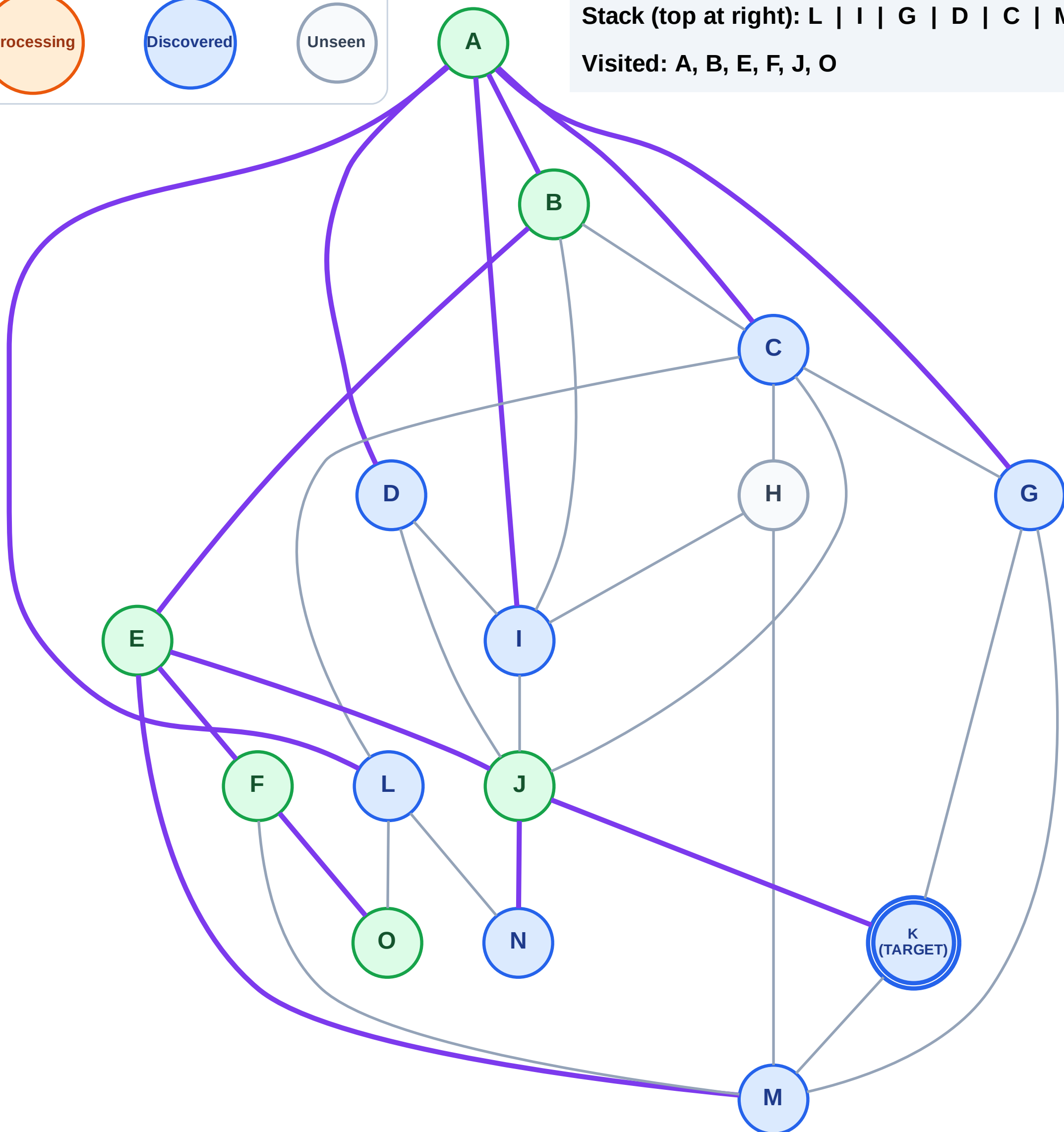
Discovered



Unseen

Stack (top at right): L | I | G | D | C | M | N | K

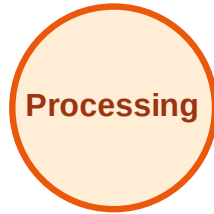
Visited: A, B, E, F, J, O



DFS Traversal

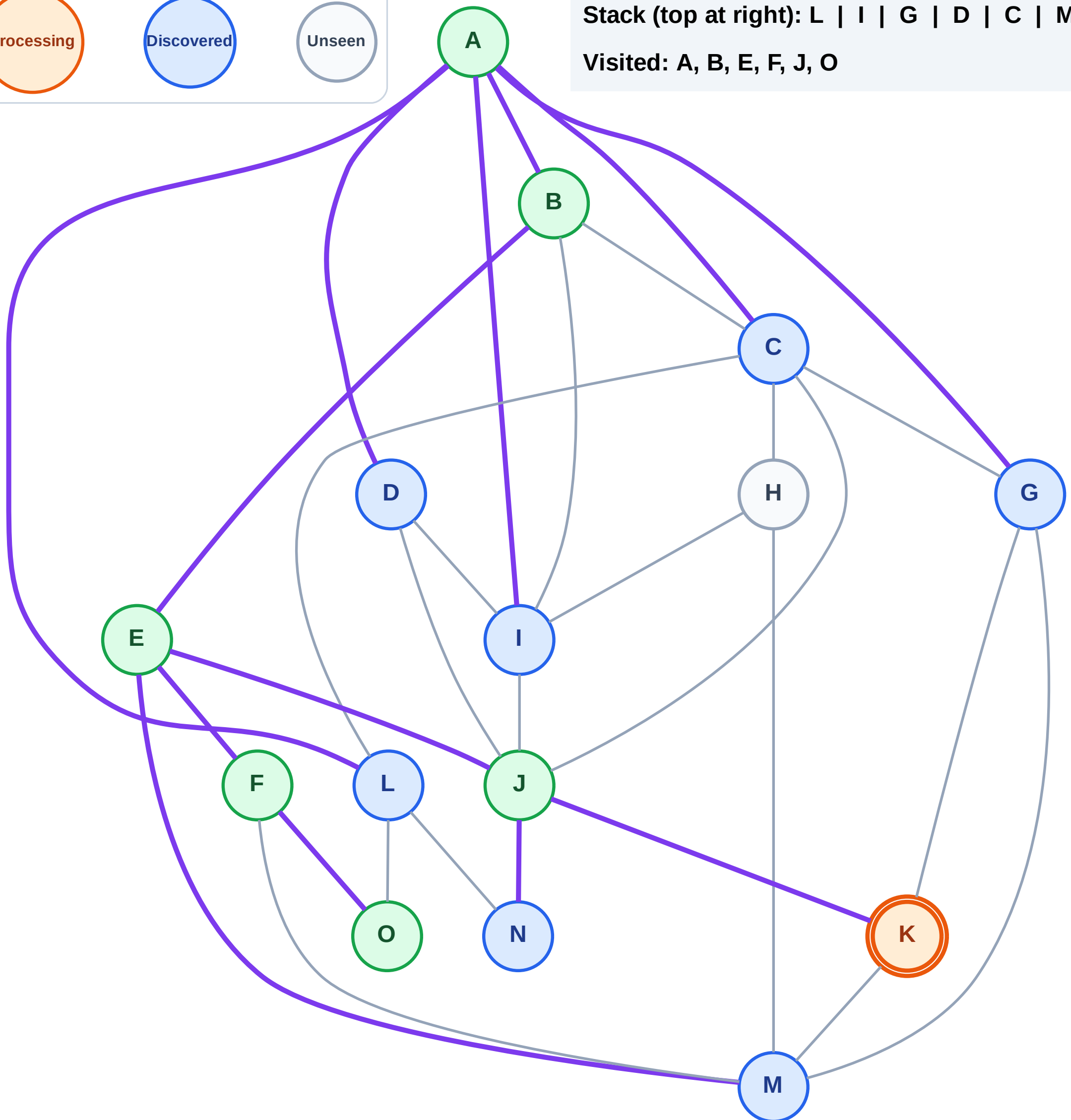
Step 14: Process node K

Legend



Stack (top at right): L | I | G | D | C | M | N

Visited: A, B, E, F, J, O



DFS Traversal

Step 15: TARGET NODE K REACHED! Path (4 hops): A → B → E → J → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): L | I | G | D | C | M | N

Visited: A, B, E, F, J, K, O

